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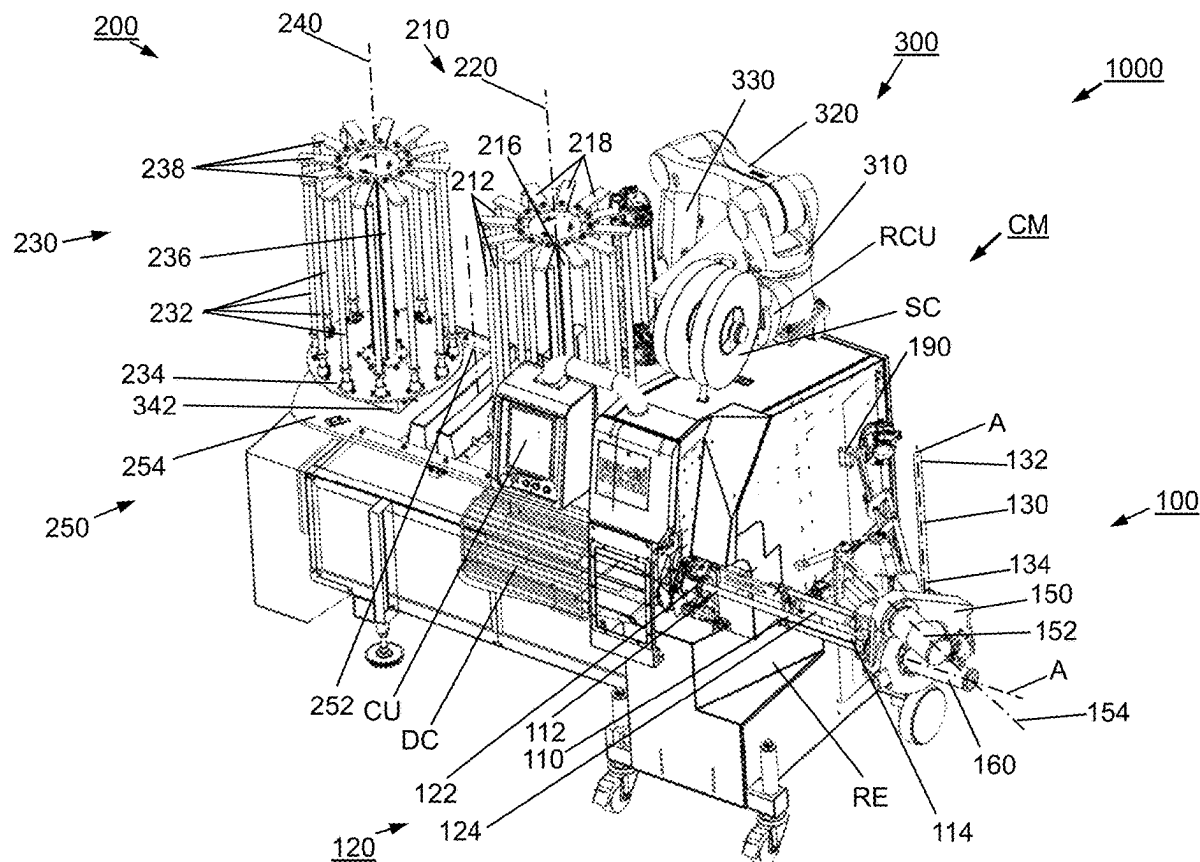
(19) **United States**(12) **Patent Application Publication**  
**Meyrahn et al.**(10) **Pub. No.: US 2025/0280840 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **SYSTEM FOR PRODUCING  
SAUSAGE-SHAPED PRODUCTS****Publication Classification**(71) Applicant: **POLY-CLIP SYSTEM GMBH & CO.  
KG**, Hattersheim (DE)(51) **Int. Cl.**  
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CPC ..... **A22C 11/0236** (2013.01); **A22C 11/0245**  
(2013.01)(21) Appl. No.: **18/862,107**(57) **ABSTRACT**(22) PCT Filed: **May 2, 2023**(86) PCT No.: **PCT/EP2023/061911**

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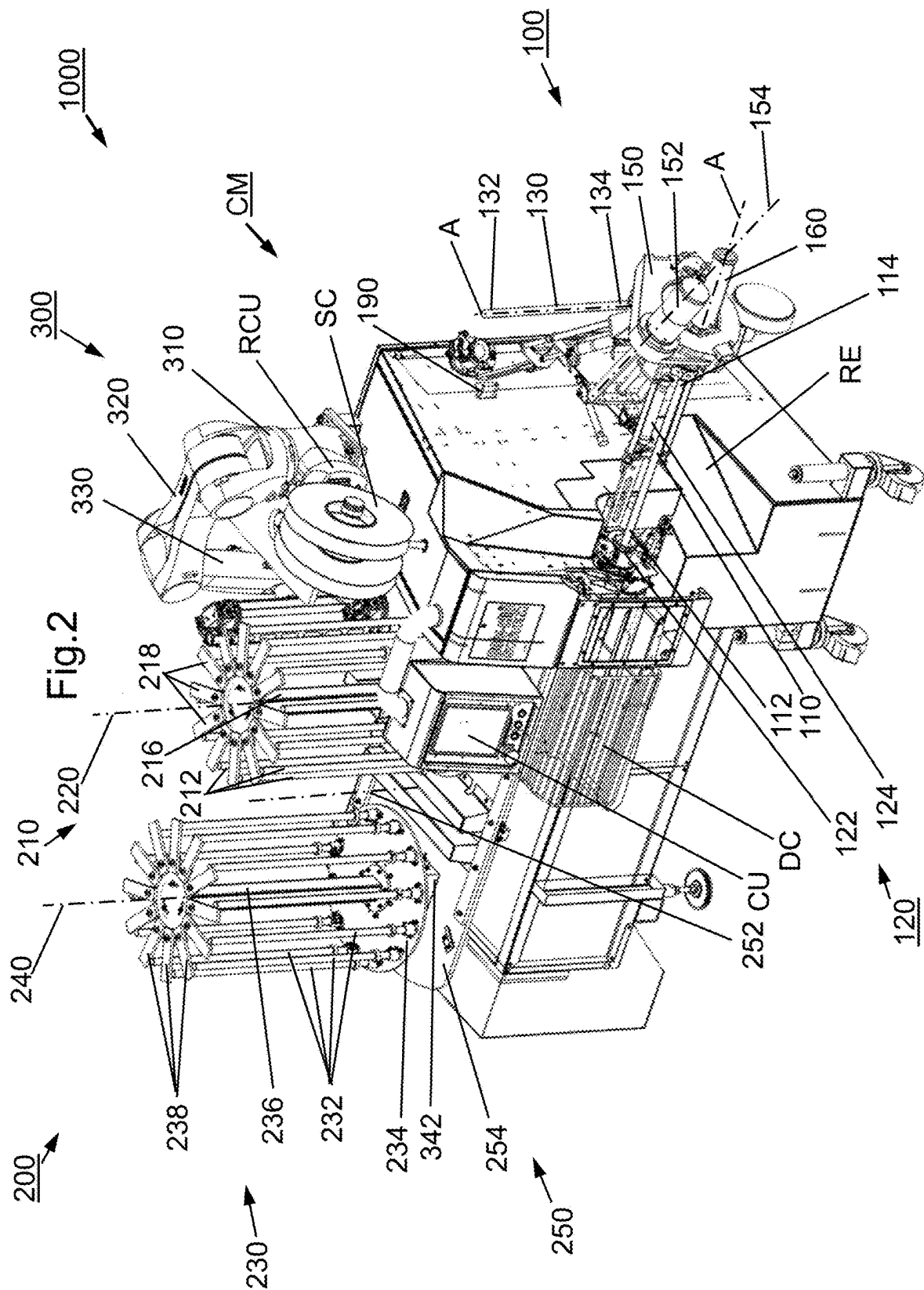
(2) Date: **Oct. 31, 2024**(30) **Foreign Application Priority Data**

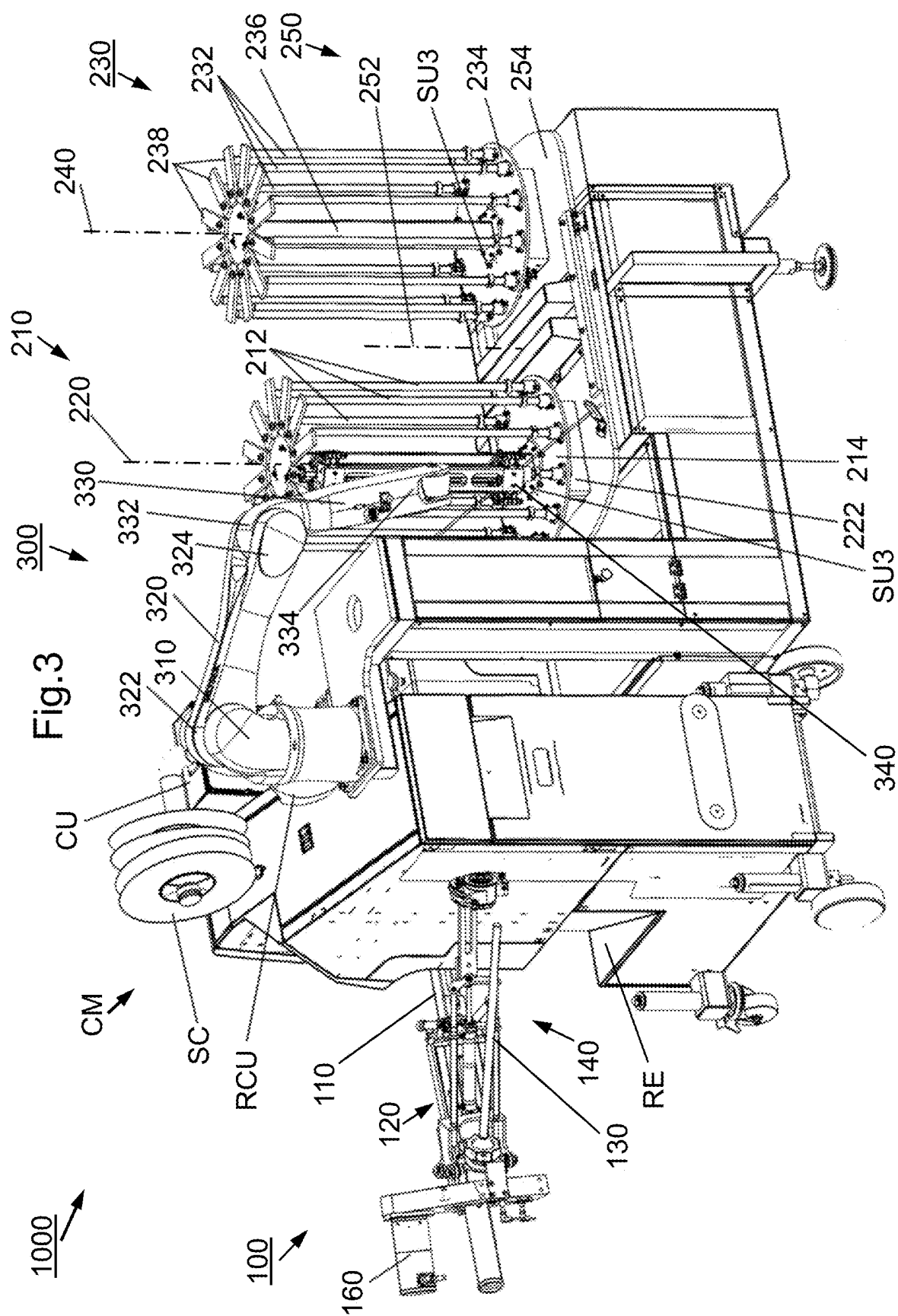
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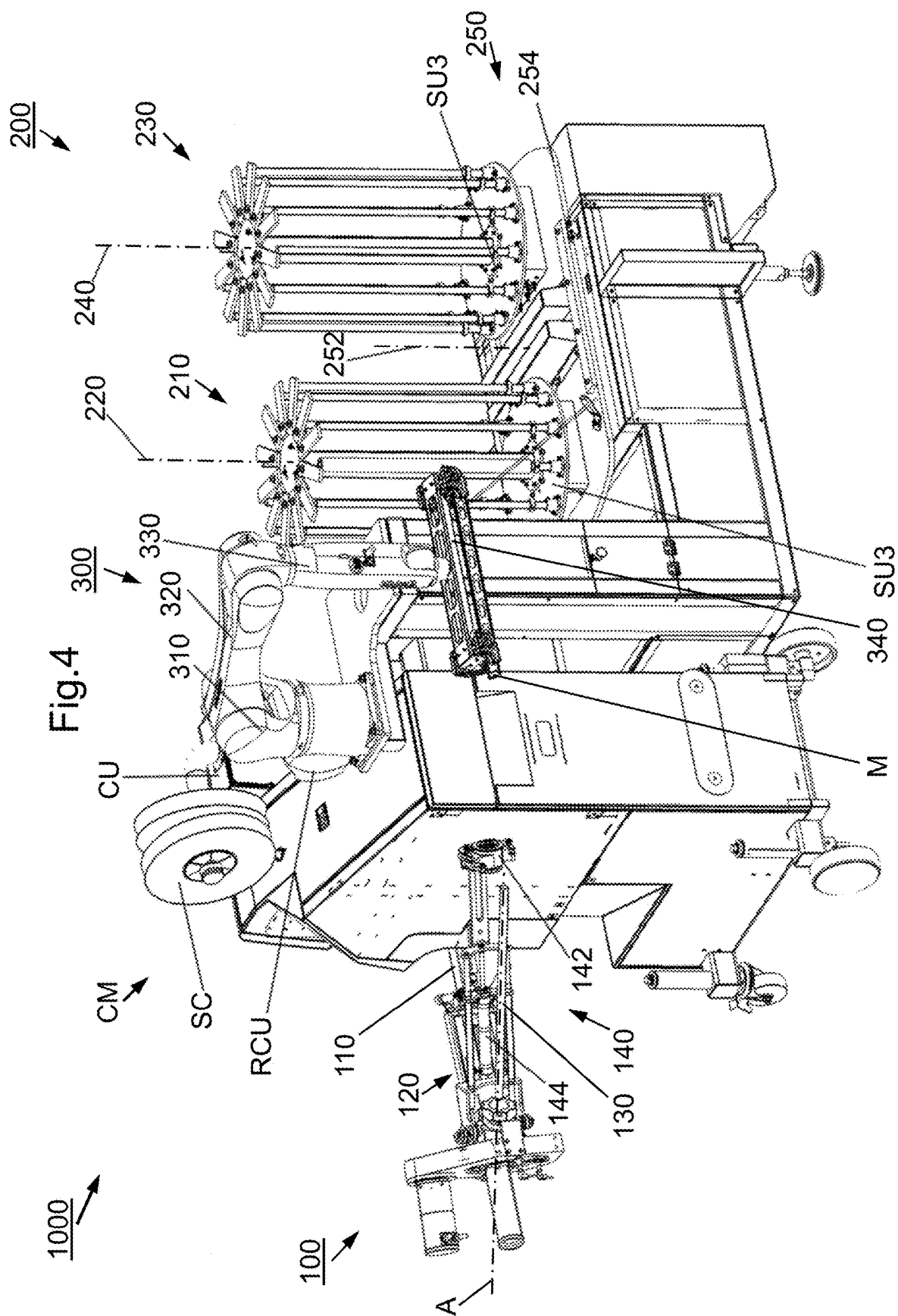
The present invention relates to system for producing sausage-shaped products, like sausages, by filling a filling material into a tubular packaging casing. The system includes a clipping machine for producing sausage-shaped products, a robotic device for transferring a tubular packaging casing supply to the clipping machine, and a storage device for storing a tubular packaging casing supply. The present invention further relates to a method for producing sausage-shaped products in the system.

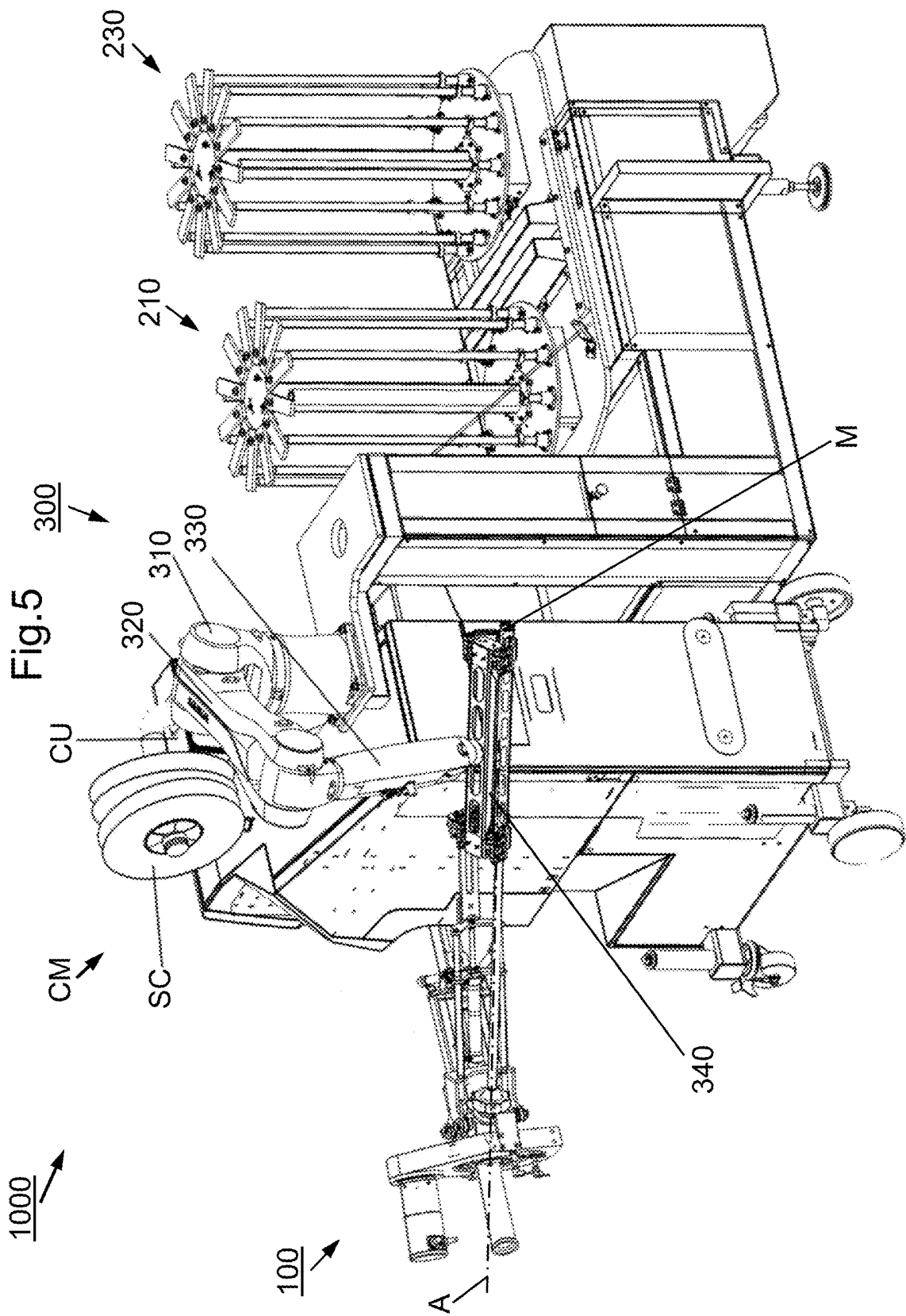


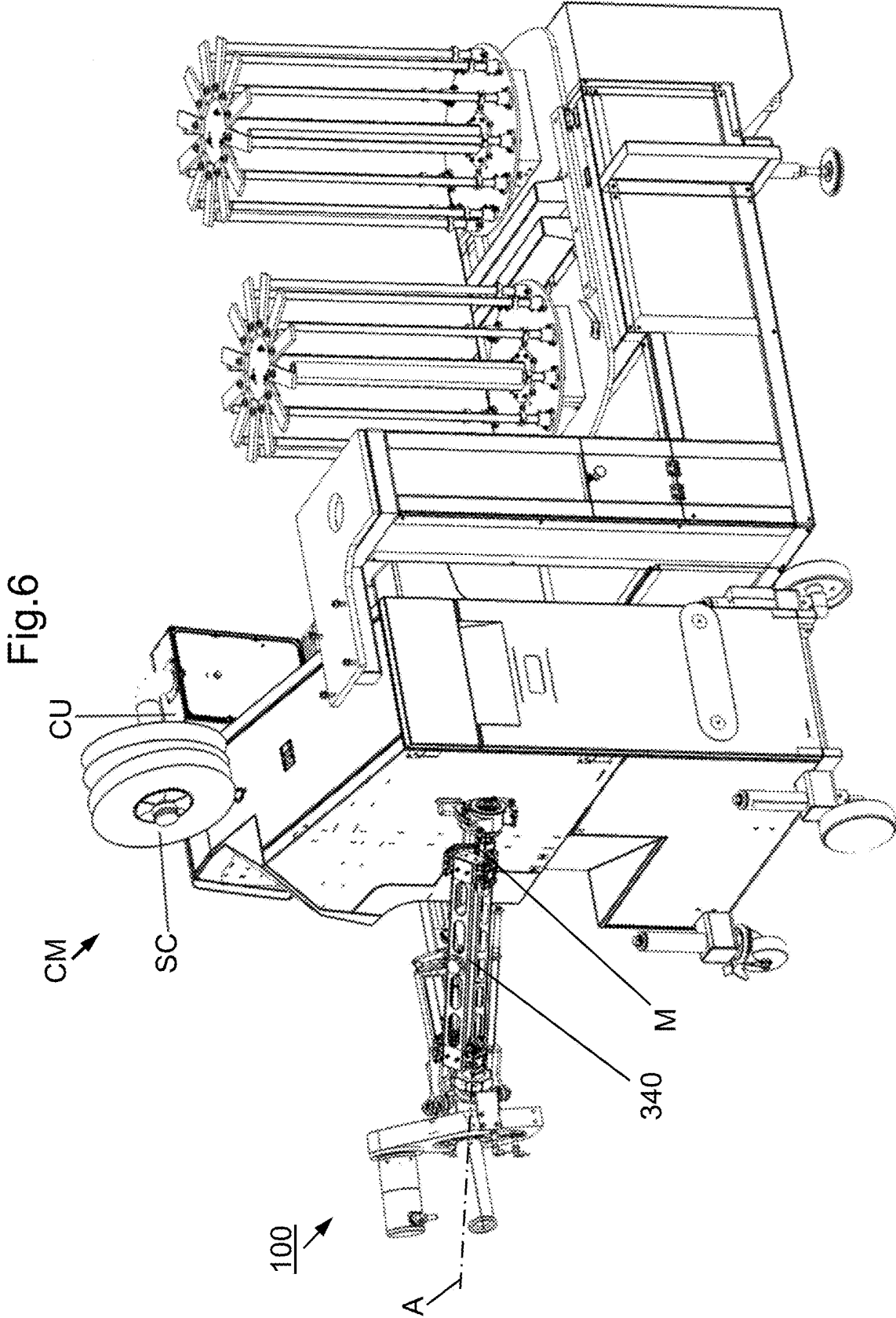












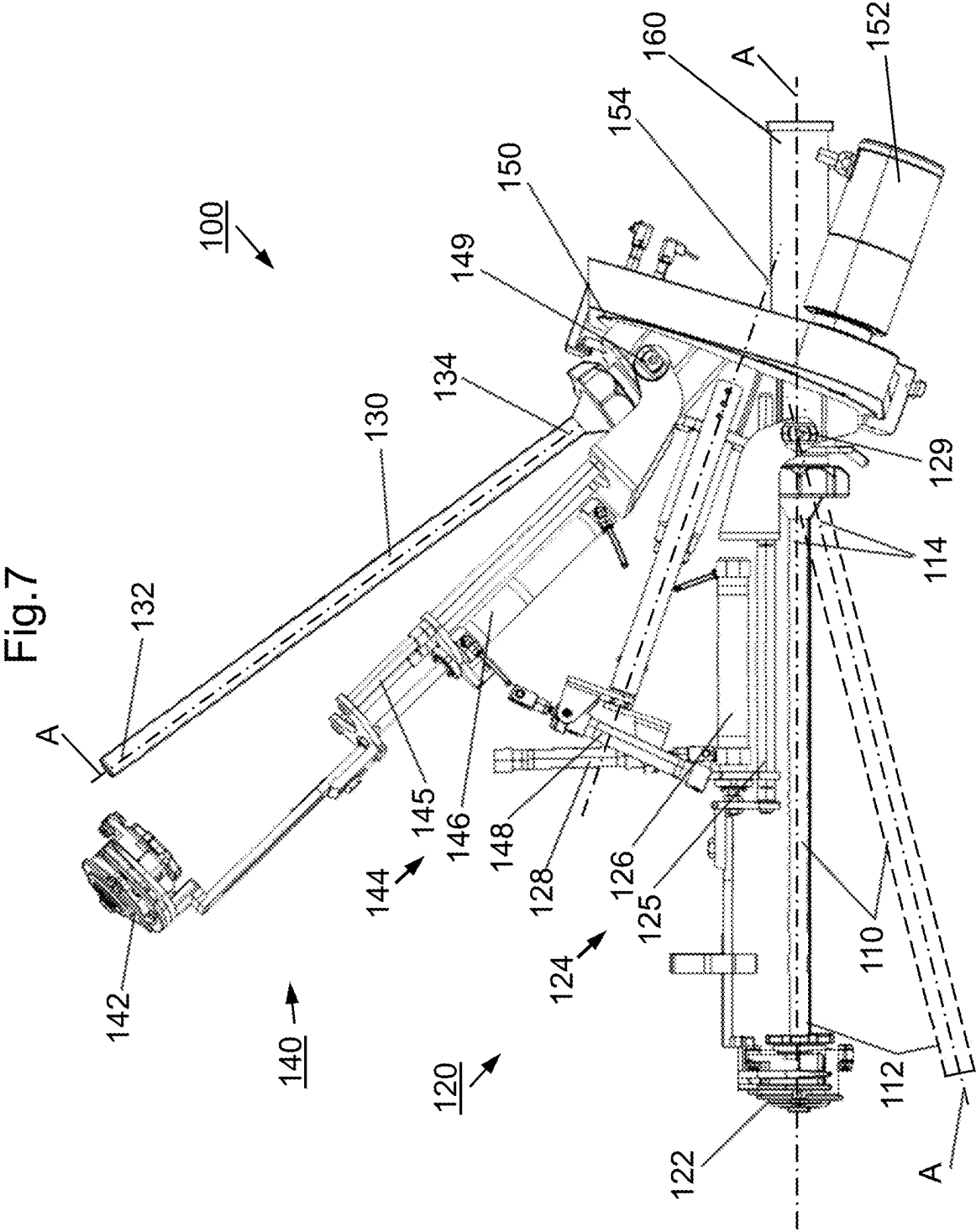
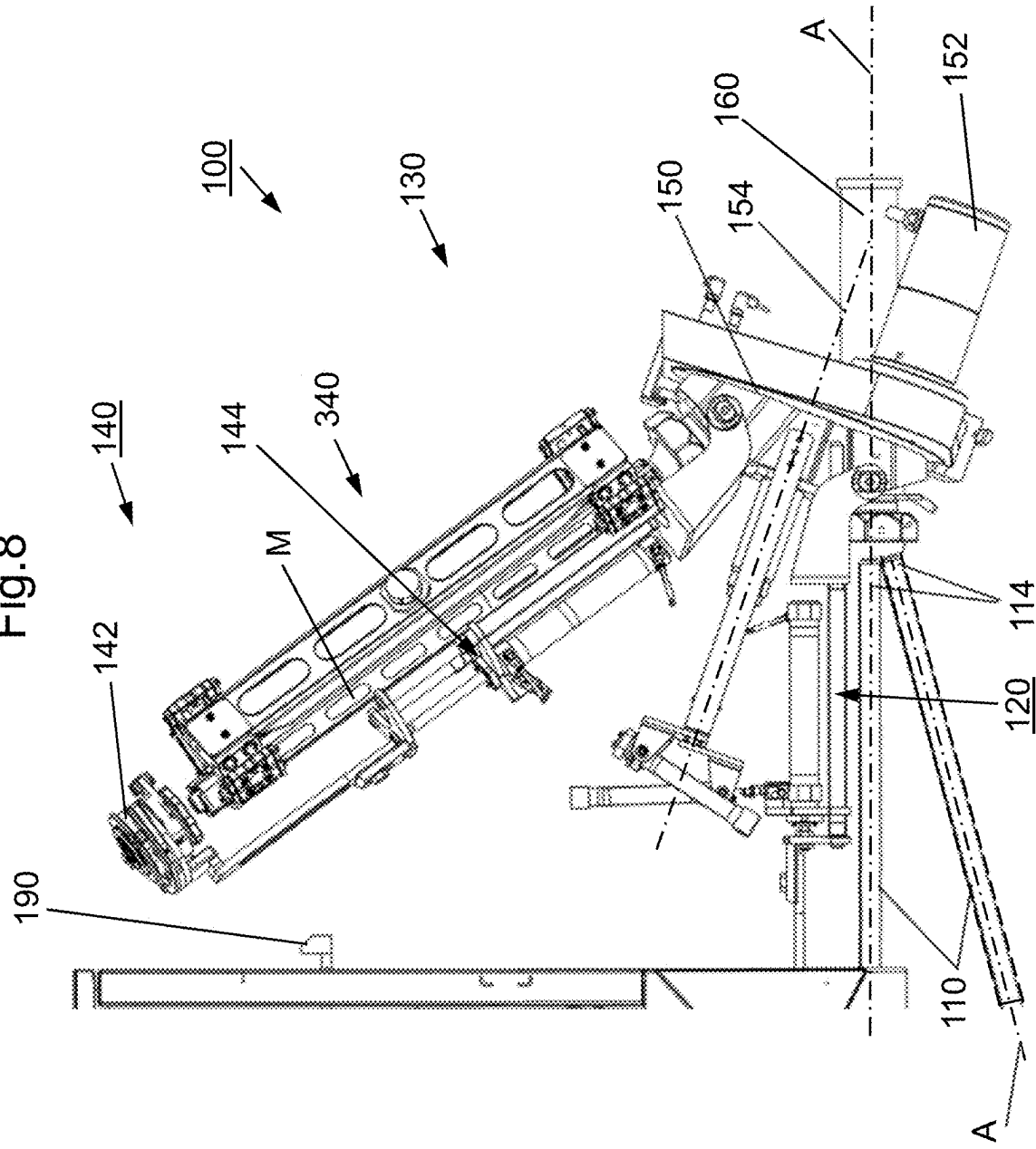
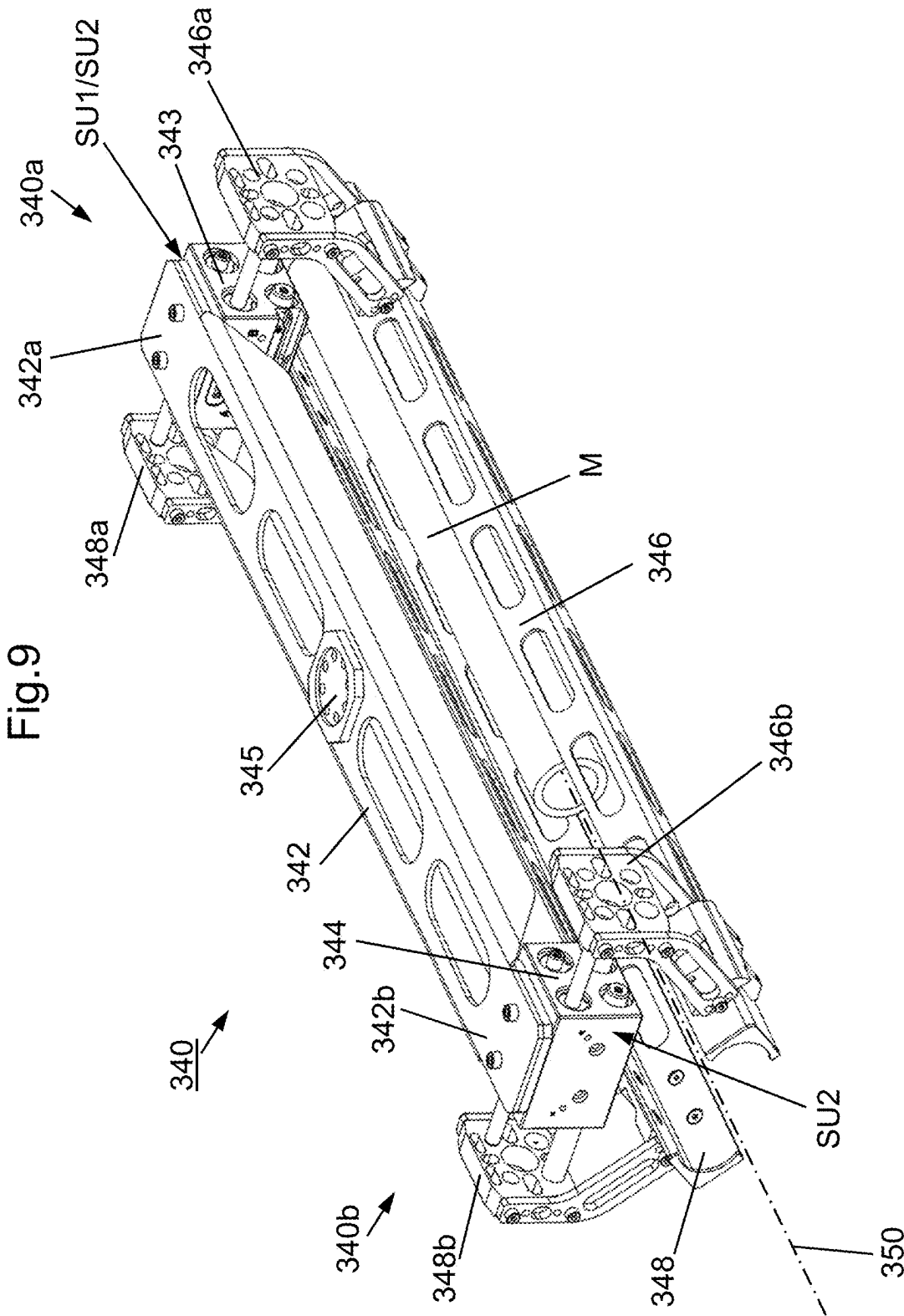
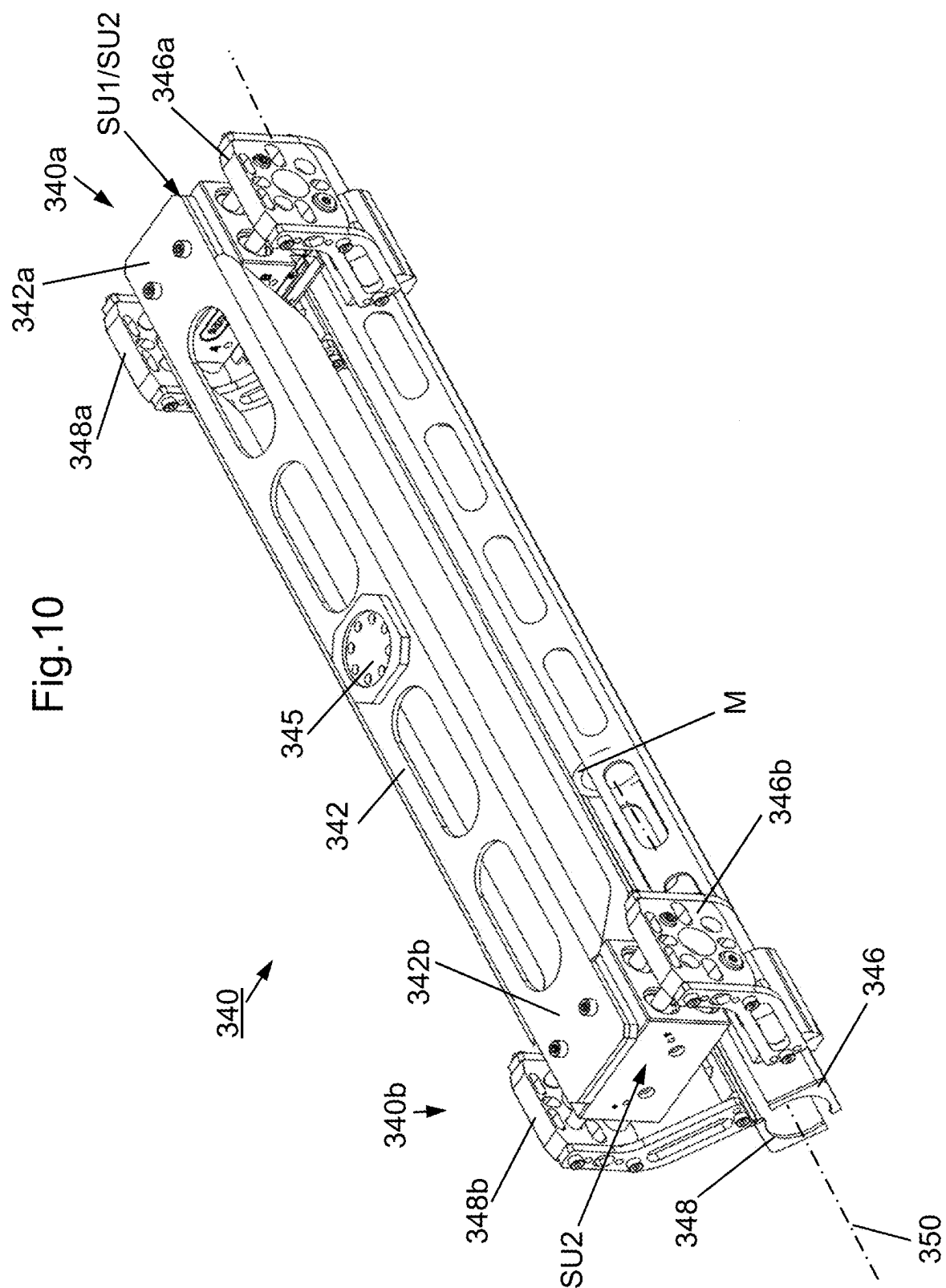




Fig.8







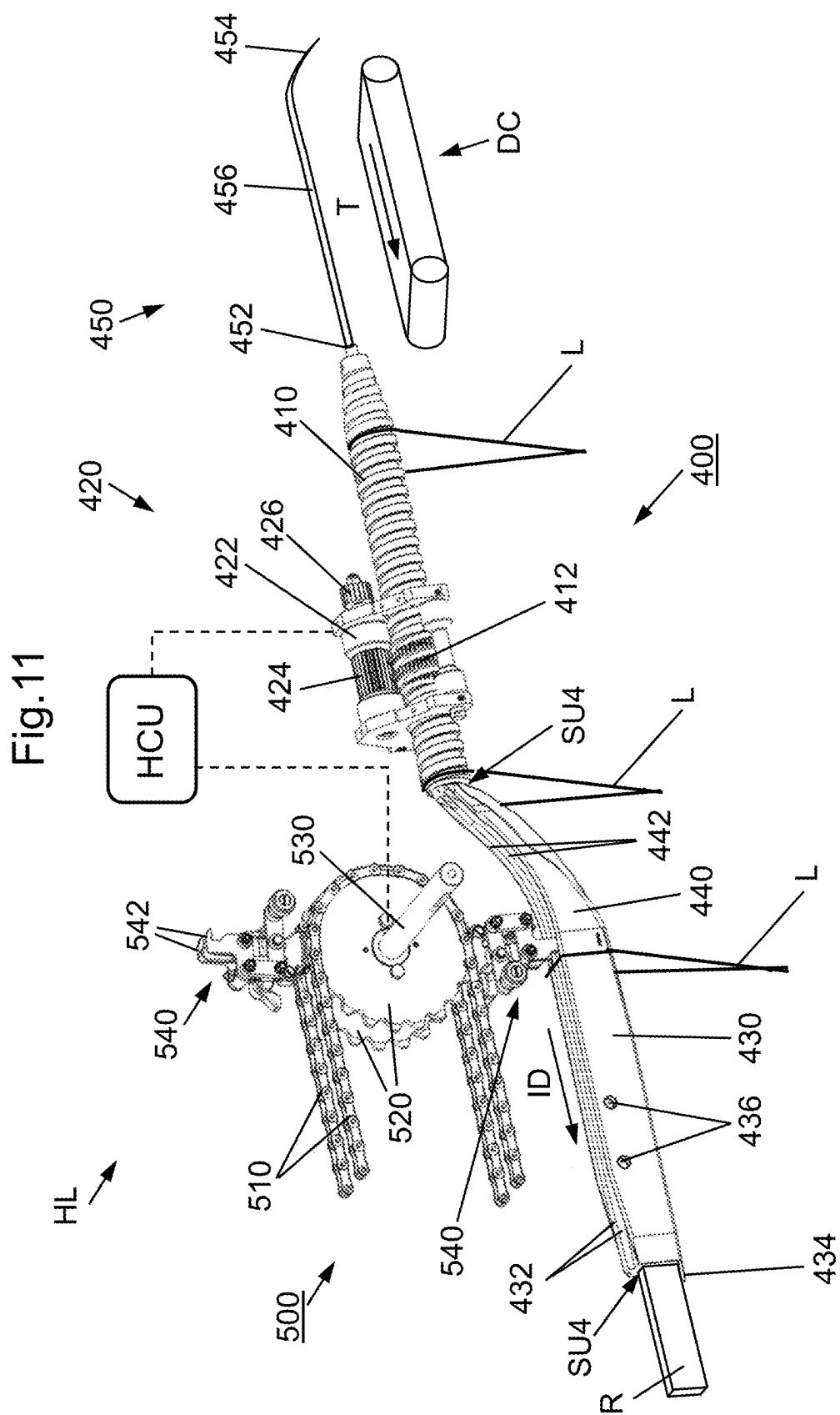
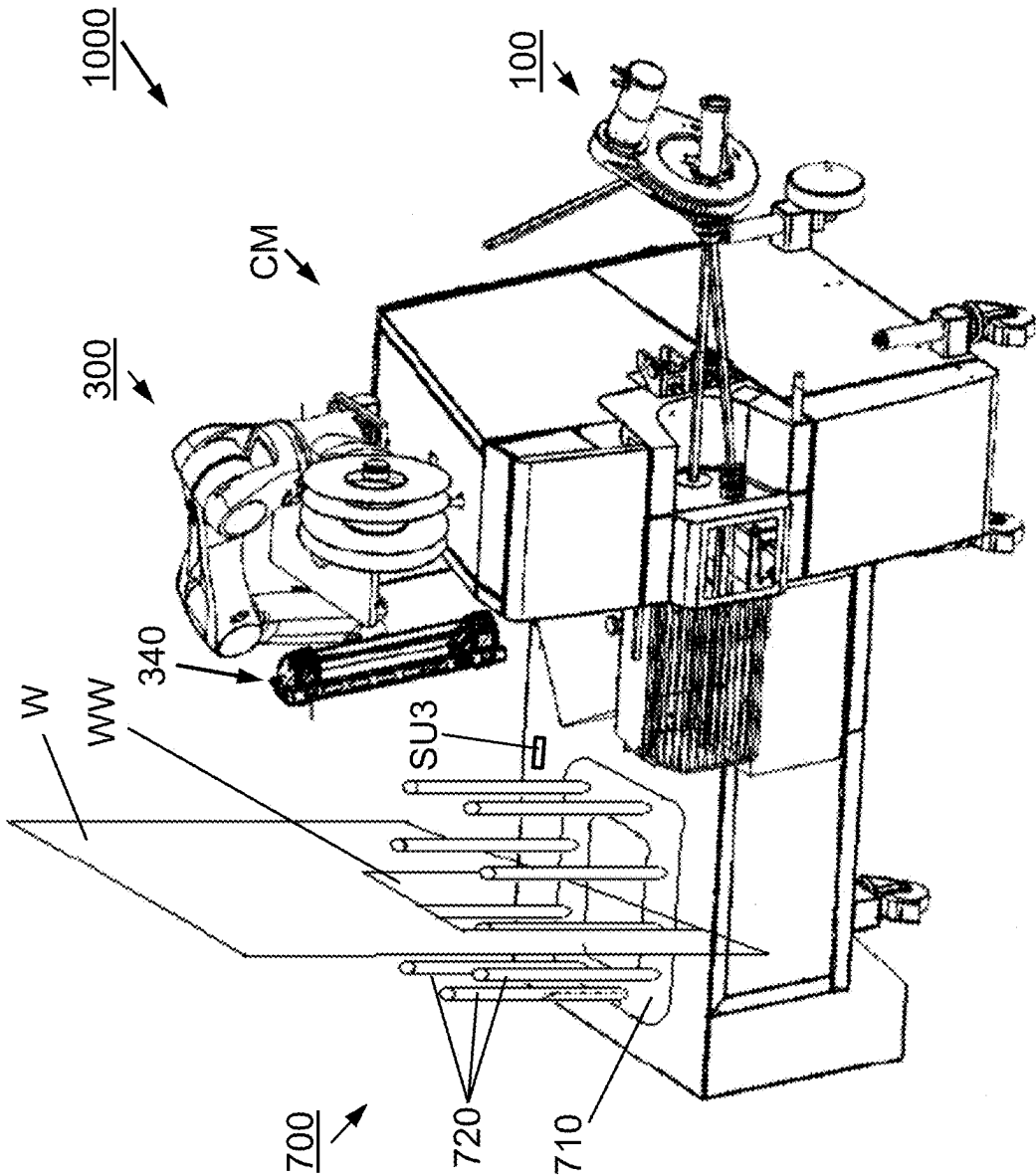


Fig.12



## SYSTEM FOR PRODUCING SAUSAGE-SHAPED PRODUCTS

**[0001]** The present invention relates to a system for producing sausage-shaped products, like sausages, by filling a filling material into a tubular packaging casing, the system includes a clipping machine for producing sausage-shaped products and a robotic device. The present invention further relates to a system for producing sausage-shaped products like sausages, by filling a filling material into a tubular packaging casing, the system includes a clipping machine for producing sausage-shaped products, a robotic device and a hanging line for suspending the sausage-shaped products on a rod-like element, like a smoking rod. Moreover, the present invention relates to a method for producing sausage-shaped products in the inventive systems.

**[0002]** In practice, packaging machines, like clipping machines, are known, for producing sausage-shaped products, like sausages, in which filling material is fed into a tubular or bag-shaped packaging casing which is stored on a filling tube, and which is closed at its front end by a closure means, like a closure clip, i.e. the end of the packaging casing pointing in the feeding direction of the filling material. After a predetermined portion of filling material has been fed into the tubular or bag-shaped packaging casing, the filled tubular or bag-shaped packaging casing is gathered by gathering means, a plait-like portion being at least substantially free from filling material is formed thereto, and at least a first closure clip is placed and closed on said plait-like portion by a clipping device, for closing the portion of the tubular or bag-shaped packaging casing just filled. Additionally, a second closure clip may be placed for closing the front end of the sausage-shaped product to be produced next. For severing the sausage-shaped product just produced from the remaining supply a tubular or bag-shaped packaging casing, a cutting device may be provided for cutting off the plait-like portion between the first and second closure clip. If it is intended to store sausage-shaped products on a rod-like element, like a smoking rod, e.g. for further treatment like cooking or smoking, a suspension element, like a suspension loop, may be provided and attached to one end of each of sausage-shaped products, or at one end of a chain of sausage-shaped products.

**[0003]** The suspension element is fed to the clipping device and fixed to the respective end of the sausage-shaped product by the closure clip that closes the respective end of the sausage-shaped product. The sausage-shaped product is then discharged from the clipping machine by a discharge device, like a belt conveyor, and fed to a hanging line, where a predefined number of sausage-shaped products is suspended by their suspension elements onto a rod-like element. The rod-like element loaded with sausage-shaped products is fed out of the hanging line and placed for further treatment e.g. in a storage frame.

**[0004]** In these known clipping machines, tubular or packaging casing, which may be provided as a supply in the form of a hollow stick of folded tubular casing material, is manually shifted onto the filling tube, by an operator. Usually, the operator pivots the filling tube out of its filling position, shifts a tubular packaging casing supply onto the filling tube, and pivots the filling tube back into its filling position. Alternatively, a tubular packaging casing supply may automatically be shifted onto the filling tube. In this

case, the tubular packaging casing supply is transferred by gravity or by a shifting element onto the filling tube held in a specific refill position.

**[0005]** Refilling a tubular packaging casing supply onto a filling tube of a clipping machine by hand is time consuming and requires the presence of an operator for each refill event. Furthermore, for automatically refilling of a tubular packaging casing supply, whereby the tubular packaging casing supply is transferred by gravity or by a shifting element onto the filling tube, requires a specific storage. Furthermore, said storage device has to be positioned relative to the filling tube in a specific position, to which the filling tube has exactly to be aligned. Moreover, a huge control amount is necessary for coordinating the operations of the components of such an arrangement. Otherwise, the tubular packaging casing supply may be damaged while being shifted onto the filling tube, which leads to production losses and increasing downtimes.

**[0006]** Thus, it is an object of the present invention, to overcome the above mentioned drawbacks, and to provide a system and a method for producing sausage-shaped products, which enable efficient production of sausage-shaped products, in which unnecessary downtimes are omitted.

**[0007]** According to the present invention, there is provided a system for producing sausage-shaped products, like sausages, by filling a filling material into a tubular packaging casing, the system includes a clipping machine for producing sausage-shaped products, a robotic device for transferring a tubular packaging casing supply to the clipping machine, and a storage device for storing a tubular packaging casing supply. The clipping machine comprises a filling tube assembly having at least a first filling tube for feeding the filling material into the tubular packaging casing stored on the first filling tube and being closed at its first end, at least a first casing brake assembly, associated with the first filling tube, for applying a braking force to the tubular packaging casing while being pulled-off from the first filling tube, and a control unit for controlling at least the clipping machine. The robotic device includes a gripper unit for gripping a tubular packaging casing supply provided by the storage device, and for transferring said tubular packaging casing supply onto the at least first filling tube of the clipping machine. The storage device includes at least a first storage assembly in which at least one tubular packaging casing supply is provided for being transferred by the robotic device onto the at least first filling tube of the clipping machine.

**[0008]** The inventive system thereby enables the production of sausage-shaped products without unnecessary downtimes, particularly caused by the process of refilling tubular packaging casing supply onto the filling tube. Furthermore, the inventive system enables a safe and reliable transfer of a tubular packaging casing supply from a storage device onto the filling tube of a clipping machine. The robotic device may be programmed and controlled on the basis of the current positions of the storage device and the first filling tube in the refill position. The moving path of the robotic device between a take-up position, in which the tubular packaging casing supply is took up by the robotic device from the storage device, and a release position, in which the tubular packaging casing supply is released onto the filling tube, may be selected according to desired parameters, like the shortest or fastest moving path, and/or dependent on the

periphery. Thus, a complex design or cumbersome alignment of the storage device and the clipping machine may be omitted.

**[0009]** It has to be understood that a clipping machine for producing sausage-shaped products by filling a filling material into a tubular packaging casing may further comprise gathering means for gathering the filled tubular packaging casing and for forming a plait-like portion thereto, the gathering means including a first displacer unit and a second displacer unit, a clipping device having a first and a second closing tool being reversibly movable between an opened position and a closed position, for applying at least one closure means to the plait-like portion and closing said closure means when the closing tools are in their closed position, and a discharge device for discharging the sausage-shaped product just produced out of the clipping machine.

**[0010]** The tubular packaging casing supply advantageously may be provided in the form of hollow sticks of folded tubular packaging casing. These hollow casing sticks may easily and reliably be gripped by the gripper unit of the robotic device, for being removed from the storage device and transferred onto the filling tube.

**[0011]** The use of a gripper unit allows the tubular packaging casing supply to be provided in various positions, like horizontal or vertical, and thus, no specific storage device is needed. Furthermore, the storage device may be designed independently from the kind of filling tube and its orientation in the refill position.

**[0012]** In an advantageous embodiment of the inventive system, the control unit of the clipping machine is adapted to at least activate the robotic device for starting the process of refilling tubular packaging casing supply onto the filling tube.

**[0013]** Since all essential information regarding the sausage-shaped product to be produced are present in the control unit of the clipping machine, an optimized time point for starting the process of refilling tubular packaging casing supply onto the filling tube of the clipping machine may be selected, such that the refill process may be shortened and unnecessary downtimes in the production process may be omitted.

**[0014]** In a further advantageous embodiment, the at least first filling tube of the clipping machine is movable at least between a filling position, in which filling material may be fed through the first filling tube into the tubular packaging casing stored on the at least first filling tube, and a refill position, in which a tubular packaging casing supply may be transferred onto the at least first filling tube.

**[0015]** Defining a refill position for the first filling tube allows an optimized operation of the robotic device, e.g. with regard to its movement path and/or the operation time.

**[0016]** For refilling tubular packaging casing supply onto the at least filling tube of the clipping machine, it is sufficient that the clipping machine includes only one filling tube. However, in order to shorten the downtime for refilling tubular packaging casing supply onto the filling tube, it is preferred that the filling tube assembly of the clipping machine includes a second filling tube and a second casing brake assembly associated with the second filling tube, wherein, preferably, the first filling tube is arranged in the filling position, while the second filling tube is arranged in the refill position.

**[0017]** In this configuration, sausage-shaped products may be produced using the first filling tube, whereas at the same

time, tubular packaging casing supply may be transferred onto the second filling tube. Accordingly, the downtime in the production of sausage-shaped products may further be shortened.

**[0018]** In a preferred design, the filling tube assembly further comprises a revolver plate, with the at least first filling tube attached to the revolver plate, the revolver plate being rotatable about a rotation axis for reversibly moving the at least the first from the filling position into the refill position.

**[0019]** In the case that the filling tube assembly includes a second filling tube, it has to be understood that also the second filling tube is attached to the revolver plate. A revolver-design of the filling tube assembly enables a quick and reliable transfer of one of the filling tubes from the filling position to the refill position, and, at the same time, the transfer of the other filling tube from the refill position into the filling position.

**[0020]** The first and second casing brake assemblies may be associated with the first and second filling tubes in different ways. The casing brake assemblies may be positioned in fixed relationship to the clipping machine in close vicinity to the filling position and the refill position, respectively, such that e.g. the casing brake assembly close to the refill position is aligned with the respective filling tube when arranged in the refill position.

**[0021]** However, in order to securely maintain the first and second filling tubes in alignment with the first and second casing brake assemblies associated therewith, the revolver plate together with the first and second filling tubes and the first and second casing brake assemblies is rotatable about a rotation axis of the revolver plate.

**[0022]** The robotic device for transferring a tubular packaging casing supply to the clipping machine, particularly for transferring a tubular packaging casing supply from the storage device onto the filling tube arranged in the refill position, may be controlled in different ways. In one embodiment, the robotic device does not include a control unit, and that the robotic device is controlled by the control unit of the clipping machine. In the system according to the present invention it is preferred that the robotic device includes a control unit, and the control unit of the clipping machine and the control unit of the robotic device may be interconnected such that the robotic device at least partially may be controlled by the control unit of the clipping machine and/or such that the clipping machine at least partially may be controlled by the robotic device.

**[0023]** According to this, the control unit of the clipping machine may generally start or stop the operation of the robotic device, whereas the control unit of the robotic device controls the process of transferring the tubular packaging casing supply from the storage device towards the refill position. Thereby the control amount of the control unit of the clipping machine may be reduced. On the other hand, it is also possible that the control unit of the robotic device may affect the operation of the clipping machine, e.g. the clipping machine may be stopped in case of missing tubular packaging casing supply to be transferred onto the filling tube in the refill position, or in case of any malfunction recognised by the robotic device.

**[0024]** In a preferred embodiment, the robotic device includes at least a first sensor unit at least for detecting the position of the gripper unit relative to the storage device

and/or the position of the gripper unit relative to the filling tube arranged in the refill position.

**[0025]** Such a sensor unit enables an exact positioning of the gripper unit for gripping the tubular packaging casing supply and an exact positioning of the tubular packaging casing supply onto the filling tube in the refill position. Thereby, damages to the tubular packaging casing supply or to components of the clipping machine, e.g. by accidental collisions, may be prevented, and production losses may be omitted. Various sensor units for detecting the position of the gripper unit may be used, like contact sensors or contactless sensors. In order to prevent damages, preferably, contactless sensors may be used, like optical sensors, inductive sensors or ultrasonic sensors.

**[0026]** In a further preferred embodiment, the robotic device includes a second sensor unit for sensing a gripping force applied to the tubular packaging casing supply by the gripper unit.

**[0027]** The sensor unit may include a suitable sensor element, like a pressure sensor, a force sensor or a piezo sensor. Thereby, the tubular packaging casing supply may securely be gripped by the gripper unit, and damages to the tubular packaging casing supply may be omitted.

**[0028]** For further optimizing the refill process, and for avoiding unnecessary downtimes, it is of advantage that the storage device includes at least one sensor unit for detecting the presence or absence of a tubular packaging casing supply in the storage device. A signal from said sensor, which indicates the presence or absence of tubular packaging casing supply in the storage device, may be sent to either the robotic device or the clipping machine, e.g. for stopping the robotic device in the case that no tubular packaging casing supply is present in the storage device. Additionally, a respective information may be sent to an operator.

**[0029]** For enabling a movement of the filling tubes from the lateral position into the refill position, when being rotated by the revolver plate of the filling tube assembly, with a reduced risk of collision, in one embodiment of the clipping machine, a recess may be provided in the housing of the clipping machine, which enables a passage of the filling tubes, and particularly, of the front ends of the filling tubes, during their movement between the lateral position and the refill position. Said recess is arranged in the region of the right edge of the front side of the clipping machine, below the filling tube when in the filling position.

**[0030]** This recess has the further advantage that the angle about which the filling tubes have to be pivoted outwards into the lateral position, i.e. the angle between the filling position and the lateral position of the filling tube, may be smaller compared with such an angle in a clipping machine, which is not provided with such a recess. Thereby, also the necessary operation space in front of the clipping machine is reduced.

**[0031]** In a preferred design, for increasing the automation degree, independently from the previously described design of the inventive system, the inventive system may further include a hanging line arranged downstream the discharge device, for suspending the sausage-shaped products discharged from the clipping machine by the discharge device, onto a rod-like element, like a smoking rod. Moreover, the hanging line may also include a control unit for controlling at least the hanging line.

**[0032]** In one embodiment, a hanging line includes, as its main components, a catching device, an infeed conveyor and a transport device.

**[0033]** The catching device, like a catching needle, is provided for catching the suspension element or suspension loop of the sausage-shaped product just produced. The catching element extends along the discharge device and into the closing region of the clipping machine. The closing region of a clipping machine is the region where the filled tubular packaging casing is gathered, a plait-like portion is formed, one or two closure clips are placed and closed on said plait-like portion, and where a suspension element is fixed thereto by one of the closure clips.

**[0034]** The suspension element, caught by the catching element, may further be guided along an infeed conveyor for feeding the suspension element with the sausage-shaped product thereon, into the hanging line. Downstream the infeed conveyor, a guide bar or sword is arranged. A rod-like element or smoking rod is positioned downstream and aligned coaxially with the guide bar, such that a suspension element or suspension loop guided along the guide bar may be transferred onto the rod-like element, which is the loading position.

**[0035]** The transport device for transferring the suspension element along the guide bar and to a desired place on the rod-like element, is arranged above the guide bar and the rod-like element in the loading position.

**[0036]** In this configuration, it is further preferred that the control unit of the hanging line is interconnected with the control unit of the robotic device and/or the control unit of the clipping machine, such that the hanging line may at least partially be controlled by the clipping machine and/or the robotic device. Accordingly, the hanging line may be started or stopped by the clipping machine and/or the robotic device, such that the production process immediately may be interrupted upon recognising e.g. the absence of rod-like elements in the hanging line or possible malfunctions of the hanging line. Thereby, production losses may be omitted. However, it is also possible that the hanging line affects the operation of the clipping machine or the robotic device, e.g. by stopping the clipping machine or the robotic device upon detection of malfunction or the absence of sausage-shaped products in the hanging line.

**[0037]** For optimizing the production process, and particularly for the suspension of sausage-shaped products onto rod-like elements, the hanging line includes at least one fourth sensor unit for detecting the presence or absence of a sausage-shaped product, a rod-like element or for detecting possible malfunction of the hanging line. Thereby, at least the operation of the hanging line may be interrupted, e.g. in the case that no sausage-shaped products to be hung up or a respective rod-like element are present. Furthermore, since the absence of sausage-shaped products in the hanging line may indicate malfunctions of the clipping machine or the robotic device, a respective information may be sent to an operator. The sensor unit may include any suitable sensor element for detecting the presence or absence of a sausage-shaped product in the hanging line, like an optical sensor element or a mechanical sensor element. Said sensor unit may be arranged at any suitable position, like in the region of the guide bar or the front end of the rod-like element. It has to be understood that the fourth sensor unit of the hanging line is coupled to the control unit of the hanging line for sending a respective signal thereto. In the case that the



control unit of the hanging line is coupled to the control unit of the clipping machine, the sensor signal may also be sent to the control unit of the clipping machine, such that the clipping machine at least partially may control the hanging line on the basis of this signal, e.g. the hanging line may be stopped by the clipping machine in case of the absence of sausage-shaped products in the hanging line.

**[0038]** According to the present invention, there is further provided method for producing sausage-shaped products in a system according to the present invention, the method comprises the steps of feeding a filling material through the filling tube of the clipping machine into a tubular packaging casing stored on the at least first filling tube, gathering the filled tubular packaging casing by the gathering means of the clipping machine and forming a plait-like portion thereto, applying at least one closure means to the plait-like portion and closing said closure means by the clipping device of the clipping machine, and discharging the sausage-shaped product just produced out of the clipping machine by the discharge device of the clipping machine. The method further comprises the steps of moving the at least first filling tube from the filling position, in which filling material is fed into the tubular packaging casing, into the refill position, in which a tubular packaging casing supply is transferred onto the at least first filling tube, and transferring by the robotic device a tubular packaging casing supply provided by a storage device onto the at least first filling tube being arranged in the refill position.

**[0039]** The inventive method further comprising the step of suspending the sausage-shaped product discharged by the discharge device on a rod-like element provided by the hanging line, and the further step of at least partially controlling by the clipping machine the robotic device and/or the hanging line.

**[0040]** The inventive method thereby provides all advantages explained in conjunction with the inventive system for producing sausage-shaped products.

**[0041]** According to the present invention, there is further disclosed a filling tube assembly of a clipping machine for producing sausage-shaped products, like sausages, by filling a filling material into a tubular packaging casing, the filling tube assembly comprising: a revolver plate, which is rotatable about a rotation axis, at least a first filling tube and a second filling tube, each having a central axis, wherein the filling tubes are mounted on the revolver plate, at least a first casing brake assembly and a second casing brake assembly, wherein the first casing brake assembly is connected to the first filling tube and the second casing brake assembly is connected to the second filling tube, and a drive device for rotating the revolver plate including the filling tubes about the rotation axis, wherein the central axis of the first filling tube and the central axis of the second filling tube define an angle larger than 0°, preferably between 15° and 75°, and more preferably between 30° and 60°.

**[0042]** Preferably, the first casing brake assembly is associated with the first filling tube and includes a first casing brake unit positioned in the region of a first end of the first filling tube, and is pivotally coupled to the revolver plate by a first telescopic casing brake holder unit, and the second casing brake assembly is associated with the second filling tube and includes a second casing brake unit positioned in the region of a first end of the second filling tube, and is pivotally coupled to the revolver plate by a second telescopic casing brake holder unit.

**[0043]** Preferably, the first casing brake unit is linearly reversibly shiftable along the first filling tube onto its first end and pivoted away from the first end of the first filling tube, and the second casing brake unit is linearly reversibly shiftable along the second filling tube onto its first end and pivoted away from the first end of the second filling tube.

**[0044]** Preferably, the first casing brake holder unit includes a telescopic guidance and a telescopic drive for extending and retracting telescopic guidance, and the second casing brake holder unit includes a telescopic guidance and a telescopic drive for extending and retracting telescopic guidance.

**[0045]** Preferably, the telescopic drives are linear drives, particularly, piston/cylinder drives.

**[0046]** Preferably, the filling tube assembly further comprises a first pivot drive unit, which is provided for pivoting the first casing brake assembly relative to the first filling tube about a pivot point at which the first casing brake assembly is coupled to the revolver plate, and the filling tube assembly further comprises a second pivot drive unit, which is provided for pivoting the second casing brake assembly relative to the second filling tube about a pivot point at which the second casing brake assembly is coupled to the revolver plate.

**[0047]** Preferably, the filling tube assembly is mounted on a framework via a joint portion.

**[0048]** Preferably, the joint portion is mounted on a side wall of the framework.

**[0049]** Preferably, the filling tube assembly is linearly and/or rotatably shiftable such that the filling tube is positioned in front of the clipping machine.

**[0050]** Preferably, a feeding socket is fixed to the revolver plate and facing away from the clipping machine, wherein feeding socket is coaxially aligned with the central axis of the filling tube.

**[0051]** According to the present invention, there is further disclosed a system for producing sausage-shaped products, like sausages, by filling a filling material into a tubular packaging casing, the system includes a clipping machine for producing sausage-shaped product, a robotic device for transferring a tubular packaging casing supply to the clipping machine, and a storage device for storing a tubular packaging casing supply and for providing the tubular packaging casing supply to the robotic device, the clipping machine comprises: a filling tube assembly as mentioned above, gathering means for gathering the filled tubular packaging casing and for forming a plait-like portion thereto, the gathering means including a first displacer unit and a second displacer unit, a clipping device having a first and a second closing tool being reversibly movable between an opened position and a closed position, for applying at least one closure means to the plait-like portion and closing said closure means when the closing tools are in their closed position, a discharge device for discharging the sausage-shaped product just produced out of the clipping machine, and a control unit for controlling at least the clipping machine and the robotic device.

**[0052]** Preferably, the robotic device includes: a gripper unit for gripping a tubular packaging casing supply provided by the storage device, and for transferring said tubular packaging casing supply onto the at least first filling tube of the clipping machine; and a control unit for controlling at least the gripper unit.

[0053] Preferably, the storage device includes: at least a first storage assembly in which at least one tubular packaging casing supply is provided for being transferred by the robotic device onto the at least first filling tube of the clipping machine.

[0054] Preferably, the first filling tube is movable between a filling position, in which filling material is fed through the first filling tube into the tubular packaging casing stored on the first filling tube, and a refill position, in which a tubular packaging casing supply is transferred onto the first filling tube.

[0055] According to the present invention, there is further disclosed a method for producing sausage-shaped products in a system as mentioned above, the method comprises the steps of: feeding a filling material through the at least first filling tube of the clipping machine into a tubular packaging casing stored on the at least first filling tube, gathering the filled tubular packaging casing by the gathering means of the clipping machine and forming a plait-like portion thereto, applying at least one closure means to the plait-like portion and closing said closure means by the clipping device of the clipping machine, discharging the sausage-shaped product just produced out of the clipping machine by the discharge device of the clipping machine, the method further comprises the steps of: moving the at least first filling tube from the filling position, in which filling material is fed into the tubular packaging casing, into the refill position, in which a tubular packaging casing supply is transferred onto the at least first filling tube, and transferring by the robotic device a tubular packaging casing supply provided by a storage device onto the at least first filling tube being arranged in the refill position.

[0056] According to the present invention, there is further disclosed a gripping device for picking up, conveying and delivering a casing tube such as a casing tube of a sausage, the gripping device comprising: a longitudinal base structure which is attached to a handling device by means of a coupling element, preferably in a non-destructively detachable manner, at least two gripper members having each at least one bottom surface and at least one top surface, wherein the gripper members of the gripping device are provided with gripping surfaces, at least two drive means, which are attached to the axial ends of the base structure, for reversibly moving the gripper members between a first position in which the gripper members are positioned on opposite sides of the casing tube to be picked up, and a second position in which the gripper members seize the casing tube to be picked up, and a control unit for simultaneously driving the drive means of the gripper members.

[0057] Preferably, the drive means are driven pneumatically, and wherein the drive means are supplied by means of a compressed air supply via a common valve and respective conduits.

[0058] Preferably, the conduits from the common valve to the respective drive means are substantially of the same length.

[0059] Preferably, a respective flow restrictor is connected to the drive means for compensating unsynchronized movements during opening and closing of the gripper members by the drive means.

[0060] Preferably, at least one of the gripper members is translatory and/or rotatory movable with respect to the base structure.

[0061] Preferably, each of the gripper members is guided by at least two guide rods.

[0062] Preferably, at least one of the gripper members comprises a detection means which is capable of detecting the contact of the casing tube, wherein preferably two opposing gripper members comprise each a detection means.

[0063] Preferably, the drive means are connected to the gripper members via at least one connecting bracket having preferably a curved shape.

[0064] Preferably, the base structure and/or the at least two gripper members comprise a plurality of through holes and oblong holes for reducing the weight of the gripping device.

[0065] According to the present invention, there is further disclosed a system for picking up, conveying and delivering a casing tube such as a casing tube of a sausage, the system comprising: a delivering bar on which the respective casing tube is stored, a receiving bar on which the respective casing tube is to be inserted, and a transporting apparatus including a handling device and a gripping device as mentioned above for conveying the casing tube from the delivering bar on the receiving bar.

[0066] Preferably, the system further comprises at least one magazine including a plurality of delivering bars, wherein the at least one magazine is mounted on a framework, wherein preferably two magazines are arranged on the framework opposite to each other.

[0067] Preferably, the framework is rotatable and configured to rotate the at least one magazine in a picking up position and a loading position, respectively.

[0068] Preferably, the at least one magazine comprises a plurality of substantially vertical delivering bars, wherein preferably the delivering bars are arranged in a circle with a predefined distance to each other.

[0069] Preferably, the at least one magazine is rotatable and configured to rotate one of the delivering bars in the picking up position.

[0070] Preferably, the gripping device of the transporting apparatus is configured to pick up a casing tube from the delivering bar and to deliver said casing tube on the receiving bar which is a filling tube of a clipping machine.

[0071] Preferably, the transporting apparatus is attached to the clipping machine, preferably on a back plate of the clipping machine or on a top plate of the clipping machine.

[0072] According to the present invention, there is further disclosed a magazine for providing at least one tubular-shaped packaging casing suitable for receiving a flowable filling material in the production of sausage-shaped products, in particular sausages, comprising: at least one mandrel for holding at least one packaging casing, wherein the at least one mandrel has a longitudinal mandrel axis, and wherein the at least one mandrel comprises a free end at a first mandrel end, and a conveyor for holding at least one mandrel in an upright position, wherein the conveyor conveys the at least one mandrel from a loading position to a delivery position.

[0073] Preferably, at least one retaining element is provided wherein the retaining element is arranged at least at the proximity of the first mandrel end for releasably retaining the packaging casing on the at least one mandrel.

[0074] Preferably, the at least one retaining element has at least one first section in which it releases the free end of the

at least one mandrel and at least one second section in which the free end of the at least one mandrel is capped by the at least one retaining element.

**[0075]** Preferably, the at least one retaining element is made of a material which is elastic at least in sections.

**[0076]** Preferably, the longitudinal mandrel axis is oriented at least approximately vertically or forms an acute angle with the vertical line.

**[0077]** Preferably, the conveyor is a circulating conveyor belt driven by a motor, wherein the conveyor belt is preferably arranged in an oval configuration.

**[0078]** Preferably, the at least one mandrel has a second mandrel end spaced apart from the first mandrel end along the longitudinal mandrel axis and is attached to a respective mandrel carrier means positioned on the conveyor.

**[0079]** Preferably, the at least one mandrel comprises at least one support element for supporting the packaging casing, the at least one support element is preferably reversibly adjustable in its position along the mandrel.

**[0080]** Preferably, the conveyor is inclined in its longitudinal axis.

**[0081]** Preferably, the at least one retaining element is held by holding means coupled by a shaft to the mandrel carrier means for supporting the holding means and transmitting forces from the mandrel carrier means to the holding means.

**[0082]** Preferably, the carrier means are holding the mandrel, the shaft and the holding means.

**[0083]** According to the present invention, there is further disclosed a production system, in particular a clipping machine, for filling and closing a tubular or bag-shaped packaging casing to produce sausage-shaped products, in particular sausages, comprising: at least one magazine as mentioned above, and a robot to pick-up a packaging casing provided at the magazine and to place it at a filling unit.

**[0084]** Preferably, the loading position and the delivery position of the conveyor is separated by a separating wall, wherein the separating wall includes at least one window for passing the conveyor including the at least one mandrel.

**[0085]** Preferably, the separating wall includes at least one sensor to monitor the at least one window of the separating wall for unexpected movements such as a hand of an operator.

**[0086]** Preferably, the at least one sensor is switched off while a mandrel is passing the window of the separating wall.

**[0087]** According to the present invention, there is further disclosed a magazine for providing at least one tubular-shaped packaging casing suitable for receiving a flowable filling material in the production of sausage-shaped products, in particular sausages, comprising: a longitudinal magazine axis, at least one mandrel for holding at least one packaging casing, wherein the at least one mandrel has a longitudinal mandrel axis, which extends at least approximately parallel to the longitudinal magazine axis, and wherein the at least one mandrel comprises a free end at a first mandrel end, and at least one retaining element arranged at least at the proximity of the first mandrel end for releasably retaining the packaging casing on the at least one mandrel.

**[0088]** Preferably, the longitudinal magazine axis is oriented at least approximately vertically.

**[0089]** Preferably, the at least one retaining element has at least one first section in which it releases the free end of the

at least one mandrel and at least one second section in which the free end of the at least one mandrel is capped by the at least one retaining element.

**[0090]** Preferably, the at least one retaining element is made of a material which is elastic at least in sections.

**[0091]** Preferably, the at least one mandrel has a second mandrel end spaced apart from the first mandrel end along the longitudinal mandrel axis and is attached to mandrel carrier means.

**[0092]** Preferably, the mandrel carrier means are rotatable around the longitudinal magazine axis.

**[0093]** Preferably, the at least one retaining element is held by holding means coupled by a shaft to the mandrel carrier means for supporting the holding means and transmitting forces from the mandrel carrier means to the holding means.

**[0094]** Preferably, the mandrel carrier means have a first diameter and the holding means have a second diameter, and wherein the first diameter is greater than the second diameter.

**[0095]** Preferably, the at least one mandrel is arranged at the mandrel carrier means along a circular ring that is perpendicular to the magazine axis.

**[0096]** Preferably, the circular ring has a ring diameter, and the first diameter is greater than the ring diameter and/or the ring diameter is larger than the second diameter.

**[0097]** Preferably, the at least one mandrel comprises at least one support element for supporting the packaging casing, the at least one support element is preferably reversibly adjustable in its position along the mandrel.

**[0098]** According to the present invention, there is further disclosed a revolver comprising at least two magazines as mentioned above, wherein the at least two magazines are arranged on revolver carrier means, which are rotatable around a revolver axis extending at least approximately parallel to the longitudinal magazine axis.

**[0099]** Preferably, the revolver carrier means have at least a first position in which one magazine is arranged in a delivery position and at least a second position in which the further magazine is arranged in a feeding position.

**[0100]** Preferably, comprising at least one separating wall for separating the first position from the second position.

**[0101]** According to the present invention, there is further disclosed a production machine, in particular a clipping machine, for filling and closing a tubular or bag-shaped packaging casing to produce sausage-shaped products, in particular sausages, comprising: at least one magazine as mentioned above and/or at least one revolver as mentioned above, and a robot to pick-up a packaging casing provided at the magazine and/or at the revolver and to place it at a filling unit.

**[0102]** According to the present invention, there is further disclosed a method for controlling a robotic device by a control unit of a clipping machine, the robotic device being arranged in a system for producing sausage-shaped products by filling a filling material into a tubular packaging casing supply arranged on a filling tube of the clipping machine, wherein the system comprises at least the clipping machine provided preferably with at least a first filling tube and preferably a second filling tube and a first and preferably a second casing brake assembly, a storage device for storing at least one tubular packaging casing supply, and a robotic device for transferring a tubular packaging casing supply from the storage device onto a filling tube of the clipping machine, wherein the control unit is adapted to provide

control commands at least to the robotic device, and wherein the robotic device includes a first gripper unit for gripping a tubular packaging casing supply and a second gripper unit for gripping the first casing brake assembly, the method comprises the steps of: removing the casing brake assembly from the filling tube by the second gripper unit; moving the robotic device into a take-up position for taking up by the first gripper unit a tubular packaging casing supply provided in a storage device; gripping the tubular packaging casing supply provided in a storage device by the first gripper unit; moving the robotic device from the take-up position toward the filling tube; transferring the tubular packaging casing supply onto the filling tube by the first gripper unit; releasing the tubular packaging casing supply onto the filling tube of the clipping machine by the first gripper unit, and positioning the casing brake assembly on the filling tube by the second gripper unit.

**[0103]** Preferably, the method further comprises the step of exchanging a first gripper unit of robotic device, for gripping a tubular packaging casing supply by a second gripper unit for gripping the at least first casing brake assembly, for reversibly removing the first casing brake assembly from the at least first filling tube.

**[0104]** Preferably, the method further comprises the step of securing the tubular packaging casing supply on the first filling tube by at least one retainer device.

**[0105]** Preferably, the robotic device includes at least a first sensor unit associated with the first gripper unit, for detecting the position of the gripper unit relative to the tubular packaging casing supply in the storage device and/or for detecting the position of the gripper unit relative to the filling tube.

**[0106]** Preferably, the first gripper unit at least a first and a second gripper element being arranged laterally along a longitudinally extending gripping axis, and at least a second sensor unit, for sensing the gripping force exerted by the at least first and second gripper elements onto the tubular packaging casing supply wherein, preferably, the method further comprising the step of: limiting the gripping force exerted by the at least first and second gripper elements onto the tubular packaging casing supply to a predefined value.

**[0107]** Preferably, the method further comprises the step of moving the first gripper unit along a predefined moving path towards the filling tube of the clipping machine.

**[0108]** Preferably, the method further comprises the step of positioning the first gripper unit with its gripping axis coaxially to the central axis of the filling tube and in front of the filling tube; and shifting the tubular packaging casing supply onto the filling tube by the first gripper unit.

**[0109]** Preferably, the method further comprises the step of detecting by the first sensor unit the position of the tubular packaging casing supply on the filling tube; and releasing the tubular packaging casing supply onto the filling tube.

**[0110]** Preferably, the robotic device is activated upon a signal indicating a specific condition of the clipping machine.

**[0111]** Preferably, the robotic device is deactivated upon a signal indicating a specific condition of the clipping machine and/or a specific condition of the storage device.

**[0112]** Preferably, the clipping machine includes a first filling tube and at least a second filling tube, with one filling tube being arranged in a filling position in which filling material may be fed into the tubular packaging casing, and the other filling tube being arranged in a refill position in

which a tubular packaging casing supply is fed onto the filling tube, and wherein the robotic device is activated upon a signal indicating one of the first or second filling tubes being arranged in or moving towards the refill position.

**[0113]** Preferably, the first gripper unit includes at least one further sensor unit for detecting the position of the gripper elements.

**[0114]** According to the present invention, there is further disclosed a system for producing sausage-shaped products, like sausages, by filling a filling material into a tubular packaging casing supply arranged on a filling tube of a clipping machine, comprising at least: a storage device for storing at least one packaging casing supply; a clipping machine provided with at least at least a first filling tube and preferably a second filling tube onto each of which a packaging casing supply can be applied; a robotic device for transferring a packaging casing supply from the storage device onto a filling tube of the clipping machine, and having at least a first gripper unit for gripping a tubular packaging casing supply, and a second gripper unit for gripping a casing brake assembly of the clipping machine; and a control unit for at least partially controlling the operation of at least the robotic device, wherein the control unit is adapted to control the operation of the robotic device depending on the filling status of a sausage-shaped product to be filled.

**[0115]** Preferably, the filling tubes of the clipping machine are configured such that one filling tube can assume a filling position provided for filling a tubular packaging casing supply with a filling material, and the other filling tube can assume a refilling position provided for loading a new tubular packaging casing supply onto the filling tube, and vice versa, wherein the control unit is configured to control the robotic device such that the robotic device applies a new tubular packaging casing supply onto the filling tube positioned in the refilling position while the other filling tube is in the filling position.

**[0116]** According to the present invention, there is further disclosed a robotic device including a gripper unit and arranged in a system for producing sausage-shaped products as mentioned above and controlled by the method as mentioned above.

**[0117]** According to the present invention, there is further disclosed a system for producing sausage-shaped products, like sausages, by filling a filling material into a tubular packaging casing, the system includes a clipping machine for producing sausage-shaped product, a robotic device for transferring a tubular packaging casing supply to the clipping machine, and a storage device for storing a tubular packaging casing supply and for providing the tubular packaging casing supply to the robotic device, the clipping machine comprises: a filling tube assembly having at least a first filling tube for feeding the filling material into the tubular packaging casing stored on the filling tube and being closed at its first end, at least a first casing brake assembly, associated with the first filling tube, for applying a braking force to the tubular packaging casing while being pulled-off from the first filling tube, and a control unit for controlling at least the clipping machine, the robotic device includes: at least a first gripper unit for gripping a tubular packaging casing supply provided by the storage device, and for transferring said tubular packaging casing supply onto the at least first filling tube of the clipping machine; and at least a second gripper unit for gripping the at least first casing brake

assembly, for reversibly removing the at least first casing brake assembly from the at least first filling tube; and the storage device includes: at least a first storage assembly in which at least one tubular packaging casing supply is provided for being transferred by the robotic device onto the at least first filling tube of the clipping machine.

**[0118]** Preferably, a receiving device is provided for at least temporarily receiving the first and/or the second gripper unit, wherein, preferably, the receiving device is provided at the clipping machine.

**[0119]** Preferably, the receiving device includes at least a first holding unit and a second holding unit for at least temporarily accommodating the first and/or second gripper unit.

**[0120]** Preferably, at least one retainer device is provided for temporarily securing the tubular packaging casing supply on the first filling tube in the refill position.

**[0121]** Preferably, the control unit of the clipping machine is adapted to at least activate the robotic device for starting the process of refilling tubular packaging casing supply onto the first filling tube.

**[0122]** Preferably, the first filling tube is movable at least between a filling position, in which filling material can be fed through the first filling tube into the tubular packaging casing stored on the first filling tube, and a refill position, in which a tubular packaging casing supply can be transferred onto the first filling tube.

**[0123]** Preferably, the filling tube assembly includes a second filling tube and a second casing brake assembly associated with the second filling tube, wherein, preferably, the first filling tube is arranged in the filling position while the second filling tube is arranged in the refill position.

**[0124]** Preferably, the filling tube assembly further comprises a revolver plate, with at least the first filling tube attached to the revolver plate, the revolver plate being rotatable about a rotation axis for reversibly moving the at least first filling tube from the filling position into the refill position.

**[0125]** Preferably, the robotic device includes a control unit RCU, and wherein the control unit of the clipping machine and the control unit of the robotic device are interconnected such that the robotic device at least partially may be controlled by the control unit of the clipping machine and/or the clipping machine at least partially may be controlled by the robotic device.

**[0126]** Preferably, the robotic device includes at least a first sensor unit at least for detecting the position of the first gripper unit relative to the storage device and/or the position of the first gripper unit relative to the first and/or second filling tube arranged in the refill position.

**[0127]** Preferably, the robotic device includes a second sensor unit for determining a gripping force applied to the tubular packaging casing supply by the first gripper unit.

**[0128]** Preferably, the storage device includes at least a third sensor unit for detecting the presence or absence of a tubular packaging casing supply in the storage device.

**[0129]** Preferably, the system further including a hanging line arranged downstream the discharge device, for suspending the sausage-shaped products discharged from the clipping machine by the discharge device onto a rod-like element, like a smoking rod, the hanging line includes a control unit for controlling at least the hanging line, wherein, preferably, the control unit of the hanging line is interconnected with the control unit of the robotic device and/or the

control unit of the clipping machine, such that the hanging line is at least partially controlled by the clipping machine and/or the robotic device.

**[0130]** Preferably, the hanging line includes at least one fourth sensor unit for detecting the presence or absence of a sausage-shaped product in the hanging line.

**[0131]** According to the present invention, there is further disclosed a method for producing sausage-shaped products in a system as mentioned above, the method comprises the steps of: producing a sausage-shaped product by a clipping machine having an at least first filling tube onto which a tubular packaging casing is stored into which filling material is fed, the method further comprises the steps of: moving the at least first filling tube from a filling position, in which filling material is fed into the tubular packaging casing, into a refill position, in which a tubular packaging casing supply is transferred onto the at least first filling tube, removing by the robotic device a first casing brake assembly from the at least first filling tube being in the refill position, and transferring by the robotic device a tubular packaging casing supply provided by a storage device onto the at least first filling tube being arranged in the refill position.

**[0132]** Preferably, the method further comprises the step of positioning by the robotic device the first casing brake assembly onto the at least first filling tube.

**[0133]** Preferably, the method further comprises the step of exchanging a first gripper unit of the robotic device, for gripping a tubular packaging casing supply by a second gripper unit for gripping the at least first casing brake assembly, for reversibly removing the first casing brake assembly from the at least first filling tube.

**[0134]** Preferably, the method further comprises the step of suspending the sausage-shaped product discharged by the discharge device on a rod-like element provided by the hanging line, wherein, preferably, the clipping machine controls at least partially the robotic device and/or the hanging line.

**[0135]** According to the present invention, there is further disclosed a method for controlling a robotic device by a control unit of a clipping machine, the robotic device being arranged in a system for producing sausage-shaped products by filling a filling material into a tubular packaging casing supply arranged on a filling tube of the clipping machine, wherein the system comprises at least the clipping machine provided with at least two filling tubes, a storage device for storing at least one tubular packaging casing supply, and a robotic device for transferring a tubular packaging casing supply from the storage device onto a filling tube of the clipping machine, wherein the control unit is adapted to provide control commands at least to the robotic device, and wherein the robotic device includes a gripper unit for gripping a tubular packaging casing supply, the method comprises the steps of: moving the robotic device into a take-up position for taking up by the gripper unit a tubular packaging casing supply provided in a storage device; gripping the tubular packaging casing supply provided in a storage device by the gripper unit; moving the robotic device from the take-up position toward the filling tube; transferring the tubular packaging casing supply onto the filling tube by the gripper unit; and releasing the tubular packaging casing supply onto the filling tube of the clipping machine by the gripper unit.

**[0136]** Preferably, the robotic device includes at least a first sensor unit associated with the gripper unit, for detect-

ing the position of the gripper unit relative to the tubular packaging casing supply in the storage device and/or for detecting the position of the gripper unit relative to the filling tube.

[0137] Preferably, the gripper unit includes at least a first and a second gripper element being arranged laterally along a longitudinally extending gripping axis, and at least a second sensor unit, for sensing the gripping force exerted by the at least first and second gripper elements onto the tubular packaging casing supply wherein, preferably, the method further comprising the step of: limiting the gripping force exerted by the at least first and second gripper elements onto the tubular packaging casing supply to a predefined value.

[0138] Preferably, the method further comprises the step of moving the gripper unit along a predefined moving path towards the filling tube of the clipping machine.

[0139] Preferably, the method further comprises the step of positioning the gripper unit with its gripping axis coaxially to the central axis of the filling tube and in front of the filling tube; and shifting the tubular packaging casing supply onto the filling tube by the gripper unit.

[0140] Preferably, the method further comprises the step of detecting by the first sensor unit the position of the tubular packaging casing supply on the filling tube; and releasing the tubular packaging casing supply onto the filling tube.

[0141] Preferably, filling tube of the clipping machine is provided with a casing brake assembly, the method further comprises the step of: positioning the casing brake assembly in front of the filling tube after the tubular packaging casing supply has been shifted onto the filling tube and prior the tubular packaging casing supply has been released onto the filling tube by the gripper unit.

[0142] Preferably, the robotic device is activated upon a signal indicating a specific condition of the clipping machine.

[0143] Preferably, the robotic device is deactivated upon a signal indicating a specific condition of the clipping machine and/or a specific condition of the storage device.

[0144] Preferably, the clipping machine includes a first filling tube and at least a second filling tube, with one filling tube being arranged in a filling position in which filling material is fed into the tubular packaging casing, and the other filling tube being arranged in a refill position in which a tubular packaging casing supply is fed onto the filling tube, and wherein the robotic device is activated upon a signal indicating one of the first or second filling tubes being arranged in or moving towards the refill position.

[0145] Preferably, the gripper unit includes at least one further sensor unit for detecting the position of the gripper elements.

[0146] According to the present invention, there is further disclosed a system for producing sausage-shaped products, like sausages, by filling a filling material into a tubular packaging casing supply arranged on a filling tube of a clipping machine, comprising at least: a storage device for storing at least one packaging casing supply; a clipping machine provided with at least two filling tubes onto each of which a packaging casing supply can be applied; a robotic device for transferring a packaging casing supply from the storage device onto a filling tube of the clipping machine; and a control unit for controlling the operation of at least the robotic device, wherein the control unit is adapted to control the operation of the robotic device depending on the filling status of a sausage-shaped product to be filled.

[0147] Preferably, the filling tubes of the clipping machine are configured such that one filling tube can assume a filling position provided for filling a tubular packaging casing supply with a filling material, and the other filling tube can assume a refilling position provided for loading a new tubular packaging casing supply onto the filling tube, and vice versa, wherein the control unit is configured to control the robotic device such that the robotic device applies a new tubular packaging casing supply onto the filling tube positioned in the refilling position while the other filling tube is in the filling position.

[0148] Preferably, the filling tubes of the clipping machine are configured such that one filling tube can assume a filling position provided for filling a tubular packaging casing supply with a filling material, and the other filling tube can assume a refilling position provided for loading a new tubular packaging casing supply onto the filling tube, and vice versa, wherein the first filling tube is provided with a first casing brake unit reversibly removable from the first filling tube and the second filling tube is provided with a second casing brake unit reversibly removable from the second filling tube, and wherein the control unit is configured to release the tubular packaging casing supply fed to the filling tube located in the refilling position when the respective casing brake unit is positioned in front of the filling tube shortly before being pushed onto the latter.

[0149] According to the present invention, there is further disclosed a robotic device including a gripper unit and arranged in a system for producing sausage-shaped products as mentioned above and controlled by the method as mentioned above.

[0150] Further advantages and preferred embodiments of the present invention will be described in the following together with the drawings listed below. The expressions “left”, “right”, “below” and “above” used in the following description, are referred to the drawings in an alignment such that the reference numbers and the notation of the figures used can be read in normal orientation.

[0151] FIG. 1: is a schematic view showing the principal design of a clipping machine for producing sausage-shaped products;

[0152] FIG. 2: is a perspective view to the front side of the system for producing sausage-shaped products according to the present invention;

[0153] FIG. 3: is a perspective view to the rear side of the system according to FIG. 2, with the robotic device in the take-up position;

[0154] FIG. 4: is a perspective view to the rear side of the system according to FIG. 2, with the robotic device in the allocation position;

[0155] FIG. 5: is a perspective view to the rear side of the system according to FIG. 2, with the gripper unit of the robotic device being coaxially arranged to the filling tube in the refill position;

[0156] FIG. 6: is a perspective view to the rear side of the system according to FIG. 2, with the gripper unit of the robotic device in the release position;

[0157] FIG. 7: is a plan view of the filling tube assembly used in the system according to the present invention;

[0158] FIG. 8: is a detailed plan view to the filling tube assembly of the system according to FIG. 2;

[0159] FIG. 9: is a perspective view to the gripper unit of the robotic device according to the present invention, in the opened position;

[0160] FIG. 10: is a perspective view to the gripper unit according to FIG. 9, in the closed position;

[0161] FIG. 11: perspective view of the main components of a hanging line used in the system according to the present invention; and

[0162] FIG. 12: is a perspective front view of the system according to the present invention, comprising a second embodiment of storage device.

[0163] The general design of a clipping machine CM for producing sausage-shaped products S, like sausages, which contain a flowable filling material in a tubular packaging casing M, is shown schematically in FIG. 1. As its main components, clipping machine CM comprises a filling tube 10 having a longitudinally and horizontally extending central axis A, with a discharge opening for discharging the filling material at its first end 12 and a feeding opening for feeding the filling material into filling tube 10 in a feeding direction F to the discharge opening of filling tube 10 at its second end 14, for example by a feeding pump of a filler. Moreover, a casing brake assembly 16 is arranged on and coaxially with filling tube 10 in the region of first end 12 of filling tube 10. Filling tube 10 is made of a suitable material, like stainless steel.

[0164] Tubular packaging casing M made of a thin sheet material is stored on filling tube 10 in a folded manner like a concertina. Tubular packaging casing M is pulled-off during the process of producing sausage-shaped products S, in particular by the feeding pressure, by filling the filling material into tubular packaging casing M. Casing brake assembly 16 provides a braking force for allowing a controlled pulling-off of tubular packaging casing M during the filling process.

[0165] Clipping machine CM further comprises gathering means 30 for gathering the filled tubular packaging casing M and forming a plait-like portion P thereto, and a clipping device 20 for closing the filled portion of tubular packaging casing M by applying closure means, like closure clips C, to plait-like portion P. These components are all arranged downstream filling tube 10.

[0166] As can be inferred from FIG. 1, clipping device 20 is positioned immediately downstream first end 12 of filling tube 10, and coaxially aligned to filling tube 10. Clipping device 20 comprises a first and a second clipping tool 22, 24 formed by a punch 22 and a die 24. It has to be noted that punch 22 and die 24 may apply and close a single closure clip C for closing just filled tubular packaging casing M, or may apply and close two closure clips C at the same time, a first closure clip C for closing the rear end of just filled tubular packaging casing M for forming a sausage-shaped product S, and a second closure clip C for closing the front end of a tubular packaging casing M subsequently to be filled.

[0167] Gathering means 30 include a first displacer unit 32 and a second displacer unit 34, wherein first displacer unit 32 is positioned downstream second displacer unit 34. First and second clipping tools 22, 24 of clipping device 20 may be positioned between first and second displacer units 32, 34, at least for applying and closing one or two closure clips C to plait-like portion P. First displacer unit 32 can reversibly be moved parallel to feeding direction F of the filling material, for forming plait-like portion P of a desired length.

[0168] If it is intended to store sausage-shaped products S on a rod-like element, like a smoking rod, e.g. for further processing, like cooking or smoking, a suspension element

L, like a suspension loop, may be provided and attached to one end of each of sausage-shaped products S, or at one end of a chain of sausage-shaped products S. Suspension element L may be fed to one of closing tools 22, 24, and may be fixed to the respective end of sausage-shaped product S by means of closure clip C that closes said end of sausage-shaped product S.

[0169] For separating sausage-shaped product S just produced from the remaining tubular packaging casing M stored on filling tube 10, a cutting device 40 may be provided. Said cutting device 40 cuts the tubular packaging casing M upstream closure clip C that closes the just filled portion of tubular packaging casing M. In case that two closure clips C have been applied to plait-like portion P by clipping device 20, cutting device 40 cuts plait-like portion P between said two closure clips C.

[0170] For discharging a sausage-shaped product S just produced from clipping machine CM in a transportation direction T being at least substantially the same as feeding direction F, a discharge device 50 is arranged downstream clipping device 20, which may be a belt conveyor comprising a conveyor belt and guide rollers.

[0171] For storing sausage-shaped products S on a rod-like element, suspension element L is caught by a catching device (not shown in FIG. 1), like a catching needle, which guides suspension element L towards the rod-like element.

[0172] Clipping machine CM further comprises a control unit CU for controlling clipping machine CM, and particularly, the components of clipping machine CM, like first and second clipping tool 22, 24 of clipping device 20, and first and second displacer unit 32, 34 of gathering means 30.

[0173] FIG. 2 is a perspective view to the front side of the system 1000 according to the present invention, for producing sausage-shaped products S.

[0174] As it can be seen in FIG. 2, system 1000 includes a clipping machine CM having a control unit CU. Clipping machine CM comprises a clipping device and gathering means, both not visible in FIG. 2, but designed as explained in conjunction with FIG. 1. Clipping machine CM further comprises a discharge device DC and a filling tube assembly 100.

[0175] As it further can be seen in FIG. 2, a supply SC of closure clips C is provided above clipping machine CM. The supply of closure clips C, or clip supply SC, includes two reels with closure clips C stored thereon in the form of chains of clips. From the clip supply SC, the chains of clips are guided towards the closing region of clipping machine CM.

[0176] Filling tube assembly 100 comprises a first filling tube 110 having a first end 112 and a second end 114, a first casing brake assembly 120 associated with first filling tube 110, a second filling tube 130 having a first end 132 and a second end 134, and a casing brake assembly 140 associated with second filling tube 130. First and second filling tubes 110, 130 are pivotally coupled to a revolver plate 150 by their second ends 114, 134 via a respective hinge (cf. FIG. 7).

[0177] First filling tube 110 faces with its first end 112 in feeding direction F towards the closing region of clipping machine CM, in which the clipping device and the gathering means are arranged. In the arrangement of FIG. 2, first filling tube 110 is in the filling position.

[0178] Second filling tube 130 faces with its first end 132 to the rear side of clipping machine CM. Second filling tube

**130** is in the refill opposition, in which tubular packaging casing supply **M** may be arranged on second filling tube **130**.

[0179] A first casing brake assembly **120** is associated with first filling tube **110** and includes a first casing brake unit **122** positioned in the region of first end **112** of first filling tube **110**, and is pivotally coupled to revolver plate **150** by a first telescopic casing brake holder unit **124**. First casing brake holder unit **124** is designed such that first casing brake unit **124** may linearly reversibly shifted along first filling tube **110** onto its first end **112** and pivoted away from first end **112** of first filling tube **110**. First casing brake holder **124** includes a telescopic guidance **125** and a telescopic drive **126** for extending and retracting telescopic guidance **125**. Telescopic drive **126** is a linear drive, particularly, a piston/cylinder drive. A pivot drive unit **128**, also in the form of a piston/cylinder drive, is provided for pivoting first casing brake assembly **120** relative to first filling tube **110** about a pivot point **129** at which first casing brake assembly **120** is coupled to revolver plate **150** (cf. FIG. 7).

[0180] A second casing brake assembly **140** includes a second casing brake unit **142** pivotally coupled to revolver plate **150** by a second telescopic casing brake holder unit **144**, for linearly reversibly shifting second casing brake assembly **140** along second filling tube **130** onto its first end **132**, and for pivoting second casing brake unit **142** away from first end **132** of second filling tube **130**. Second casing brake holder unit **144** is of similar design as first casing brake holder unit **124**, and includes a telescopic guidance **145** and a telescopic drive **146** for extending and retracting telescopic guidance **145**. Telescopic drive **146** is a piston/cylinder drive. A pivot drive unit **148**, also in the form of a piston/cylinder drive, is provided for pivoting second casing brake assembly **140** relative to second filling tube **130** about a pivot point **149** at which second casing brake assembly **140** is coupled to revolver plate **150** (see also FIG. 7).

[0181] In FIG. 7, telescopic guidance **125** of first casing brake holder **124** is in the retracted position, in which first casing brake unit **122** is positioned on first end **112** of first filling tube **110**, whereas telescopic guidance **145** of second casing brake holder **144** is in the extended position, in which second casing brake unit **142** is moved away from first end **132** of second filling tube **130**.

[0182] The pivot movement of filling tubes **110**, **130** and casing brake assemblies **120**, **140** is executed in a common plane, which is at least approximately with respect to clipping machine **CM**.

[0183] Filling tube assembly **100** further includes a drive device **152** including an electric motor and a suitable transmission or gear unit for rotating revolver plate **150** about a rotation axis **154**. Rotation axis **154** and filling tubes **110**, **130** are arranged in a common plane that is aligned at least approximately horizontally, when one of filling tubes **110**; **130** is in the refill position and the respective other filling tube **130**; **110** is in the filling position. Rotation axis is arranged in an acute angle relative to axis **A** of filling tube **110** in the filling position and between axes **A** of filling tubes **110**, **130**.

[0184] Filling tube **110** positioned in the filling position may pivoted laterally outwardly in the common plane of rotary axis **154**, from the filling position as shown in FIGS. 2, 7 and 8 into a lateral position as indicated in dotted lines in FIGS. 7 and 8. In the lateral position and the refill

position, filling tubes **110**, **130** are arranged at least approximately equiangular to rotation axis **154**.

[0185] As it further can be seen in FIG. 2, filling tube assembly **100** is provided with a feeding socket **160** fixed to revolver plate **150** and facing a way from clipping machine **CM**. Feeding socket **160** is coaxially aligned with central axis **A** of filling tube **130** when in the filling position (cf. FIGS. 7, 8). A feeding device, like a filler, for feeding filling material, like sausage meat, to clipping machine **CM**, may be coupled via feeding socket **160** to clipping machine **CM**.

[0186] As explained in conjunction with the principal design of clipping machine **CM** shown in FIG. 1, tubular packaging casing supply **M** is made of a thin sheet of tubular material that is folded like a concertina. Tubular packaging casing supply **M** thereby forms a hollow stick of folded tubular casing, which may be transferred onto filling tube **110**, **130**. Tubular packaging casing **M** may further be closed at one end by a closure clip **C**, which forms the front end of the first sausage-shaped product **S** to be produced with this tubular packaging casing supply **M**. For handling tubular packaging casing supply **M**, it is of advantage that said tubular packaging casing supply **M** has an inherent stability, for preventing tubular packaging casing supply **M** from unintentional unfolding, and that is sufficient for transferring tubular packaging casing supply **M** at least from a storage device onto filling tube **110**, **130** of clipping machine **CM**.

[0187] System **1000** for producing sausage-shaped products **S** further comprises a storage device **200** for storing a plurality of tubular packaging casing supplies **M** to be loaded onto first and second filling tubes **110**, **130** when arranged in the refill position. Storage device **200** as shown in FIGS. 2 and 3 comprises a first storage assembly **210** and a second storage assembly **230**.

[0188] First storage assembly **210** includes a plurality of vertically arranged receiving pins **212** of equal length, that are fixed on the upper side of a circular rotation plate **214** by their lower ends, and on a circle concentrically to the center of rotation plate **214** (cf. FIG. 3). A central post **216** is mounted to rotation plate **214** with its lower end, and is coaxially aligned to a rotation axis **220** of rotation plate **214**. Rotation axis **220** extends vertically to rotation plate **214** and through its center. At the upper end of central post **216**, retaining elements **218** are arranged. Retaining elements **218** are made of an elastic material, like rubber, have an approximately rectangular shape and extend radially outwardly from central post **216** towards the upper ends of receiving pins **212**, such that their outer ends are laid up with pretension on the upper ends of receiving pins **212**. Storage device **210** further includes a rotary drive **222** (cf. FIG. 3) coupled to the lower side of rotation plate **214**, for rotating first storage assembly **210** about rotation axis **220**.

[0189] Second storage assembly **230** is of identical design as first storage assembly **210**, and includes a rotation plate **234** with a vertical rotation axis **240** and a plurality of receiving pins **232** arranged vertically with their lower ends attached to rotation plate **234**, and on a circle concentrically to the center of rotation plate **234**. A central post **236** is arranged at the center of rotation plate **234** and coaxially aligned to rotation axis **240** of rotation plate **234** and through its center. At the upper end of central post **236**, retaining elements **238** are arranged. Retaining elements **238** that are also made of an elastic material, like rubber, have a rectangular shape and extend radially outwardly towards the upper ends of receiving pins **312**, and are laid up with pretension



on the upper ends of receiving pins 312. Second storage assembly 230 is rotatable about rotation axis 240 by a rotary drive 242 mounted to the lower side of rotation plate 234.

[0190] In FIG. 2, first storage assembly 210 is positioned in an unloading position, in which tubular packaging casing supply M is unloaded from first storage assembly 210 to be transferred onto second filling tube 130 arranged in the refill position, and storage assembly 230 is in a loading position, in which tubular packaging casing supply M is loaded onto storage assembly 230.

[0191] Storage device 200 further includes a rotation assembly 250 for rotating first and second storage assemblies 210, 230 about a rotation axis 252. First and second storage assemblies 210, 230 are mounted on a common rotation table 254 on the same height and with rotation axes 220, 240 are aligned parallel to each other and to rotation axis 252. Rotation table 254 of rotation assembly 250 is rotated about rotation axis 252 by a rotation mechanism arranged below rotation table 254.

[0192] System 1000 for producing sausage-shaped products S further comprises a robotic device 300 having a control unit RCU for controlling the movement of robotic device 300. Robotic device 300 has a base 310, a first arm 320 having a first end 322 and a second end 324. First arm 320 is pivotally coupled to base 310 by its first end 322. Robotic device 300 further has a second arm 330 with a first end 332 and a second end 334. Second arm 330 is pivotally coupled to second end 324 of first arm 320 by its first end 332 (cf. FIG. 3). Base 310 further includes a rotary drive mechanism for being rotated about a vertical rotation axis together with first and second arms 320, 330 relative to clipping machine CM.

[0193] As further can be seen in FIG. 2, clipping machine CM is provided with a recess RE, which enables the passage of filling tubes 110, 130, and particularly their first ends 112, 132, during their movement from the lateral position into the refill position, when being rotated by revolver plate 150 of filling tube assembly 100. Recess RE is provided in the form of a cut-out in the housing of the clipping machine, in the region of the right edge of the front side of the clipping machine CM, below the filling tube 110, 130 when in the filling position.

[0194] Recess RE provides the advantage that the angle about which filling tubes 110, 130 have to be pivoted outwards into the lateral position, i.e. the angle between the filling position and the lateral position of filling tubes 110, 130, may be smaller compared with such an angle in a clipping machine, which is not provided with such a recess RE. Thereby the danger of a collision between filling tubes 110, 130 and the housing of clipping machine CM is reduced. Moreover, the reduction of the angle about which filling tubes 110, 130 have to be pivoted from the filling position into the lateral position may also reduce the necessary operation space in front of clipping machine CM.

[0195] At the right side surface of clipping machine CM, an abutment unit 190 is arranged and positioned such that second casing brake assembly 140 abuts against abutment unit 190, when pivoted away from second filling tube 130, as shown in

[0196] FIG. 2. Abutment unit 190, which can be seen in FIG. 8 in more detail, may further include a sensor unit for detecting second casing brake assembly 140, or first casing

brake assembly 120, respectively, in this position, in which tubular packaging casing supply M may be transferred onto second filling tube 130.

[0197] Moreover, clipping machine CM is provided with a recess RE, which enables the passage of filling tubes 110, 130, and particularly their first ends 112, 132, during their movement from the lateral position into the refill position, when being rotated by revolver plate 150 of filling tube assembly 100 (cf. FIGS. 3 and 4). Recess RE is provided in the form of a cut-out in the housing of the clipping machine, in the region of the right edge of the front side of the clipping machine, below the filling tube when in the filling position.

[0198] Recess RE provides the advantage that the angle about which filling tubes 110, 130 have to be pivoted outwards into the lateral position, i.e. the angle between the filling position and the lateral position of filling tubes 110, 130, may be smaller compared with such an angle in a clipping machine, which is not provided with such a recess RE. Thereby the danger of a collision between filling tubes 110, 130 and the housing of clipping machine CM is reduced. Moreover, the reduction of the angle about which filling tubes 110, 130 have to be pivoted from the filling position into the lateral position may also reduce the necessary operation space in front of clipping machine CM.

[0199] As particularly can be seen in FIG. 3, which is a perspective view to the rear side of the system according to FIG. 2, a gripper unit 340 for gripping tubular packaging casing supply M is mounted at second end 334 of second arm 330. In FIG. 3, gripper unit 340 is in the take-up position, in which tubular packaging casing supply M provided by first storage assembly 210 and held ready on a receiving pin 212. Said receiving pin 212, from which tubular packaging casing supply M is taken up by gripper unit 340, is arranged in the provision position. The design and function of gripper unit 340 will be explained in detail in conjunction with FIGS. 9 and 10 below.

[0200] Control unit RCU of robotic device 300 according to the embodiment of FIG. 2, is mounted to base 310 of robotic device 300. Robotic device 300 is mounted to the framework of clipping machine CM by its base 310. Alternatively, a separate framework for robotic device 300 may be provided, which is arranged in a defined position relative to clipping machine CM, or which is fixedly coupled to the framework of clipping machine CM.

[0201] FIG. 4 shows robotic device 300 in an allocation position. In this position, tubular packaging casing supply M is removed from first storage device 210 and held by gripper unit 340 of robotic device 300 approximately horizontally and centrally behind clipping machine CM, to be transferred next onto second filling tube 130 being arranged in the refill position.

[0202] FIG. 5 is a perspective view to the rear side of the system 1000 with gripper unit 340 of robotic device 300 being coaxially arranged to central axis A of second filling tube 130 in the refill position, and onto which tubular packaging casing supply M has to be transferred.

[0203] FIG. 6 shows robotic device 300, and particularly, gripper unit 340 in the release position, in which tubular packaging casing supply M is positioned on filling tube 130 by gripper unit 340. For clarity reasons, in FIG. 6, only gripper unit 340 of robotic device 300 is shown.

[0204] From the position shown in FIG. 5, tubular packaging casing supply M is moved linearly along central axis A of filling tube 130 and shifted thereon by robotic device

**300**, which thereby moves into the release position shown in FIG. 6. In the release position, gripper unit **640** is moved from its closed position (cf. FIG. 9) in which tubular packaging casing supply **M** is grasped, into its opened position (cf. FIG. 10), for releasing tubular packaging casing supply **M** onto filling tube **130**.

[0205] FIG. 8 is a plan view to filling tube assembly **100** of clipping machine **CM** according to FIG. 6, with gripper unit **340** in the release position, and immediately prior releasing tubular packaging casing supply **M** onto filling tube **130**.

[0206] In the situation shown in FIG. 6, second casing brake assembly **140** is pivoted away from second filling tube **130**, such that gripper unit **340** together with tubular packaging casing supply **M** may access second filling tube **130** (cf. FIG. 7).

[0207] Thereafter, as can be seen in FIG. 8, second casing brake assembly **140** is pivoted towards second filling tube **130**, such that second casing brake unit **142** is positioned in front of first end **132** of second filling tube **130**. Second casing brake holder unit **144** is in the extended position. In this position of second casing brake assembly **140**, gripper unit **340** is moved into its open position, for releasing tubular packaging casing supply **M** onto second filling tube **130**.

[0208] FIG. 9 is a perspective view to gripper unit **340** of robotic device **300** in the opened position.

[0209] Gripper unit **340** has a first end **340a** and a second end **340b**. Gripper unit **340** comprises a longitudinally extending, essentially rectangular base plate **342** with a first end **342a** that coincides with first end **340a** of gripper unit **340**, and a second end **342b** that coincides with second end **340b** of gripper unit **340**. On the lower side of base plate **342** (according to FIG. 9), and in the regions of its first end **342a**, a first drive assembly **343** is arranged that includes two piston/cylinder drives with the pistons extending in a plane parallel to the plane of base plate **342** and at least approximately vertical to its longitudinal extension. A second drive assembly **344**, which is identically designed to first drive assembly **343**, is mounted to base plate **342** in the regions of its second end **342b**, and with the pistons of the piston/cylinder drives extending at least approximately vertical to the longitudinal extension of base plate **340** and in the same plane as the pistons of first drive assembly **343**. At the outer ends of the pistons of the piston/cylinder drives of first and second drive assemblies **343**, **344**, brackets **346a**, **346b**, **348a**, **348b** are mounted. Brackets **346a**, **346b**, **348a**, **348b** are angled and fixed to one end to the pistons of the piston/cylinder drives such that their other ends face downwards from base plate **342**, and brackets **346a**, **348a**; **346b**, **348b** at first and second ends **342a**, **342b** of base plate **342** are directed to each other.

[0210] Gripper unit **340** is provided with a first gripper element **346** and a second gripper element **348**, each in the form of an angled bar having an approximately V-shaped cross section and a length corresponding to the length of base plate **342**. First and second gripper elements **346**, **348** are mounted to brackets **346a**, **348a**; **346b**, **348b** such that they are aligned parallel to each other and with the open side of the “V” facing to each other. First and second gripper elements **346**, **348** may reversibly be moved by first and second drive assemblies **343**, **344**, between an opened position, as shown in FIG. 9, and a closed position, as shown in FIG. 10. First and second gripper elements **346**, **348** are

aligned parallel and equidistant to a central axis or gripping axis **350** there between. In the closed position, a tubular packaging casing supply **M** is grasped by first and second gripper elements **346**, **348** and held coaxially to gripping axis **350**.

[0211] System **100** according to the present invention may further include a hanging line **HL** for suspending sausage-shaped products **S** produced in clipping machine **CM** and discharged from clipping machine **CM** by discharge device **DC**, on a rod-like element **R**, like a smoking rod.

[0212] FIG. 11 is a perspective view of the main components of hanging line **HL**. Hanging line **HL** includes amongst others, a control unit **HCU**, an infeed device **400** and a transport device **500**. In FIG. 11, only the conveyor unit of discharge device **DC** is shown, which in this case is a belt conveyor for conveying sausage-shaped products **S** out of clipping machine **CM**.

[0213] Infeed device **400** includes an actively driven infeed conveyor in the form a spindle **410** with an outer surface at least similar to a screw thread that is supported by a spindle bearing **420**, a guide bar **430** and a connecting element **440** connecting the downstream end of spindle **410** with the upstream end of guide bar **430**. Connecting element **440** has the shape of a segment of a circle, at least at its upper surface.

[0214] The upstream end of connecting element **440** has an approximately circular cross-section adapted to the cross-section of screwed spindle **410**, to allow a sliding transfer of a suspension element **L** from screwed spindle **410** onto connecting element **440**. The downstream end of connecting element **440** has an at least approximately rectangular shape adapted to the cross-sectional shape of guide bar **430**.

[0215] In the upper surface of connecting element **440**, two parallel grooves **442** are arranged, which extend in an infeed direction **ID**. Also, in the upper surface of guide bar **430**, two parallel grooves **432** are arranged, which extend in infeed direction **ID**, and which are aligned with grooves **442** in connecting element **440**.

[0216] A portion of screwed spindle **410** has a toothed surface **412** which is engaged by a toothed portion **424** of a support roller **422** of spindle bearing **420**. At a free end of support roller **422**, a toothed wheel **426** is arranged, via which support roller **422** may be rotated, for driving screwed spindle **410** at a desired speed. As a drive for infeed device **400**, an electric motor or the like may be provided.

[0217] For supporting guide bar **430**, holes **436** are provided in both side surfaces of guide bar **430**, into which indexing pins of an indexing device may engage. At its upstream end, guide bar **430** is fixedly coupled to connecting element **440**. The upstream end of connecting element **440** is radially supported by screwed spindle **410**. Connecting element **440** has an engaging element, like a pin, at its upstream end that extends into the downstream end of spindle **410**, in which it is rotatably supported, e.g. in a roller bearing, for enabling a rotational movement of screwed spindle **410** relative to connecting element **440**. Pairs of oppositely arranged indexing pins which may engage holes **436** in guide bar **430** provide radial and rotational support for guide bar **430**, and enable passage of suspension elements **L** along guide bar **430**, by pairwise alternately engaging and disengaging holes **436** in guide bar **430**.

[0218] A support element **434** extends from the downstream end guide bar **430**, for supporting a rod-like element **R** to be loaded with sausage-shaped products **S**. The other

end of rod-like element R (not shown) is also supported so that rod-like element R is held horizontally aligned at its two ends during loading sausage-shaped products S thereon. A suspension element L guided along guide bar **430** may thereby be transferred onto said rod-like element R. The upstream end of rod-like element R is supported on support element **434**, such that rod-like element R in the loading position shown in FIG. **11**, and guide bar **430** are arranged approximately coaxially.

[0219] In FIG. **11**, for clarity reasons, only an upstream portion of rod-like element R and the suspension elements L of the sausage-shaped products S are shown, with the suspension elements L representing the sausage-shaped products S to be transported along infeed device **400**.

[0220] As further shown in FIG. **11**, a longitudinally extending catching device **450**, like a catching needle, is attached to the upstream end of spindle **410** and extends coaxially thereto into the counter direction of infeed direction ID and along discharge device DC. Catching device **450** has a first end **452**, with which catching device **450** is mounted to spindle **410**, and a second end **454**, which extends into the closing region of clipping machine CM for catching suspension element L of sausage-shaped product S just produced, and a central portion **456** extending between first and second ends **452**, **454**. Second end **454** is curved downwardly into the closing region of clipping machine CM. Central portion **456** of catching device **450** is a straight portion and is aligned at least approximately parallel to transportation direction T of the conveyor unit of discharge device DC. A suspension element L caught by catching device **450** is guided by catching device **450** to spindle **410** while the body of sausage-shaped product S laying on the conveyor unit of discharge device DC, is conveyed out of clipping machine CM.

[0221] A further component of hanging line HL is transport device **500** for transporting the sausage-shaped products S fed into hanging line HL by infeed device **400**, along guide bar **430** and rod-like element R, and for placing said sausage-shaped products S on rod-like element R.

[0222] Transport device **500** is arranged vertically above guide bar **430** and rod-like element R in the loading position, and extends in infeed direction ID. Transport device **500** according to FIG. **11** is embodied as a chain conveyor forming the conveyor means having two endless conveyor chains **510** which are arranged parallel to each other, and with each conveyor chain **510** being wound about a first and a second deflection roller **520**. In FIG. **2**, only first deflection rollers **520** are shown, which are arranged on a common axle **530**. First deflection rollers **520** form the upstream end of horizontally arranged transport device **500**. The (not shown) downstream end of transport device **500** is of identical constitution, and includes two second deflection rollers about which conveyor chains **510** are wound, and which are also arranged on a common axle.

[0223] Transport device **500**, and particularly conveyor chains **510** are actively driven by one of its common axles **530** of the deflection rollers **520** via, for example, an electric motor. Transport device **500** is driven such that the lower run of conveyor chains **510** moves in infeed direction ID.

[0224] Transportation elements **540** are arranged in regular intervals on conveyor chains **510**. Each transportation element **540** includes a pair of hooks **542**, with each hook **542** engaging one of grooves **432**, **442** of guide bar **430** and

connecting element **440** for engaging a suspension element L of a sausage-shaped product S to be suspended on a rod-like element R.

[0225] Hanging line HL according to FIG. **11** is further provided with a fourth sensor unit SU4 arranged in the region of support element **434**. In FIG. **11**, only possible positions of fourth sensor unit SU4 are indicated.

[0226] Fourth sensor unit SU4 may detect the presence or absence of a rod-like element R in the loading position. Alternatively or additionally, the same fourth sensor unit SU4, or an additional sensor unit, may also detect the presence of a sausage-shaped product S while being passed by a transportation element **540**. It has to be noted that, for detecting the presence of a sausage-shaped product S at an earlier time point, fourth sensor unit SU4 may also be arranged upstream transport device **500**, e.g. in the region of the downstream end of spindle **410**.

[0227] Fourth sensor unit SU4 is coupled to control unit HCU, which controls the operation of hanging line HL, like the operation of infeed device **400** or transport device **500**. Accordingly, based on a signal received from fourth sensor unit SU4, control unit HCU may control e.g. transport device **500**, which may be stopped in the case that fourth sensor unit SU4 has detected the absence of a sausage-shaped product, or control unit HCU may control the drive device of spindle **410**, e.g. such that the speed of spindle **410** is increased or decreased, in order to control the distance or time between two sausage-shaped products being transferred from spindle **410** to guide bar **430**.

[0228] Furthermore, control unit HCU of hanging line HL is coupled to control unit CU of clipping machine CM. Thus, the signal from fourth sensor unit SU4 may also be sent to control unit CU, such that also control unit CU of clipping machine CM may affect the operation of hanging line HL. Hanging line HL may completely be stopped by control unit CM of clipping machine CM, e.g. in case of the absence of sausage-shaped products S or rod-like elements R, or in case of any malfunction of hanging line HL. Moreover, the general operation of hanging line HL, like starting or stopping hanging line HL during the production process, may be controlled by clipping machine CM. Control unit HCU of hanging line HL may alternatively or additionally be connected to control unit RCU of robotic device **300**, such that also robotic device **300** may at least partially control hanging line HL, like stopping hanging line HL when a malfunction of robotic device has been detected.

[0229] FIG. **12** is a perspective front view of system **1000** according to the present invention, comprising a second embodiment of storage device **200**.

[0230] System **100** according to FIG. **12** is of similar design as system **100** of FIG. **2**, and differs only in the design of storage device **700**.

[0231] The second embodiment of storage device **700** is arranged at a table **710** formed by a plane of the frame of system **100** and replaces storage device **200**.

[0232] Storage device **700** comprises a conveyor **710** for holding at least one mandrel or receiving pin **720**, wherein the conveyor **710** conveys the at least one mandrel or receiving pin **720** from a provision position, which is the same position as the provision position explained in conjunction with storage device **200**, to a loading position arranged at least approximately opposite to the provision position. Conveyor **710** includes an at least approximately horizontally circulating conveyor belt **712** driven by a motor

(not shown), wherein conveyor belt **712** is arranged in an approximately oval configuration.

**[0233]** Also storage device **700** includes a separating wall partition wall **W**, which is arranged between the provision position and the loading position of storage device **700**. Partition wall thereby separates the operation range of robotic device **300** from the operation range of an operator or a further robotic device.

**[0234]** For enabling passage of receiving pins **720**, partition wall **W** has at least one window **WW**, the size of which corresponds to the space needed for conveyor **710** and mandrels or receiving pins **72** to pass there through. It has to be noted that more than one window **WW** may be provided in partition wall **W**. Two windows **WW** may be provided such that each conveyor run, the forward run moving towards the provision position, and the back run moving backwards to the loading position, has its own window **WW**, whereby the total open surface in partition wall **W** may be reduced.

**[0235]** For producing sausage-shaped products **S** in system **1000**, tubular packaging casing **M**, stored on first filling tube **110** in the form of tubular packaging casing supply **M** and closed at its front end by a closure clip **C**, is filled with filling material which is fed through first filling tube **110** of filling tube assembly **100** into tubular packaging casing **M** in feeding direction **F**. Thereby, tubular packaging casing **M** is pulled off from first filling tube **110** by the feeding pressure of the filling material. First casing brake assembly **120** exerts a braking force onto tubular packaging casing **M** while being pulled off from first filling tube **110**. After a predetermined portion of filling material is fed into tubular packaging casing **M**, filled tubular packaging casing **M** is gathered and a plait-like portion is formed to the gathered portion of tubular casing **M** by the gathering means. Two closure clips **C** are applied to the plait-like portion and closed by the clipping device, a first closure clip **C** for closing the just filled tubular packaging casing **M** for forming a sausage-shaped product **S** and a second closure clip **C** for closing the front end of sausage-shaped product **S** subsequently to be produced. A suspension element **L** may be provided and attached to one end of sausage-shaped product **S** by means of one of a closure clips **C**. For severing the just produced sausage-shaped product **S** from the remaining tubular packaging casing supply **M**, the plait-like portion is cut between the two closure clips **C**, and the sausage-shaped product **S** just produced may be discharged from clipping machine **CM** by discharge device **DC** in transportation direction **T**.

**[0236]** Suspension element **L** of sausage-shaped product **S** just produced, is caught by catching device **DC** and guided towards infeed device **400** of hanging line **HL**. After sausage-shaped product **S**, and more particular, the body of sausage-shaped product **S**, has left the conveyor unit of discharge device **DC**, sausage-shaped product **S** is transferred by spindle **410** of infeed device **400** via connecting element **440** to guide bar **430**. At the left end of guide bar **430**, suspension element **L** is caught by hooks **542** of one of transportation elements **540**, and sausage-shaped product **S** is guided along guide element **430** and rod-like element **R** to a desired position on rod-like element **R**. Hooks **542** are guided in grooves **432** of guide bar **430**, such that they engage suspension element **L**. At the desired position on rod-like element **R**, where sausage-shaped product **S** has to be placed, hooks **542** are pivoted (counter clockwise in FIG. 7) to release suspension elements **L**, and thus, sausage-

shaped products **S**, on the desired position on rod-like element **R**. Transport device **500** thereby subsequently positions sausage-shaped products **S** on rod-like element **R**, starting from its left end towards its right end, until a predefined number of sausage-shaped products **S** is stored on rod-like element **R**, and with predefined distances there between.

**[0237]** After the tubular packaging casing supply **M** on first filling tube **110** is used up, first filling tube **110** is pivoted from the filling position (cf. FIG. 2) into the lateral position (indicated in FIGS. 7 and 8). Filling tube assembly **100** is then rotated about rotation axis **154** of revolver plate **150** and about an angle of at least approximately 180°. During this rotational movement of filling tube assembly **100**, first filling tube **110** is transferred into the refill position and second filling tube **130**, with a tubular packaging casing supply **M** stored thereon, is moved into the lateral position. Thereafter, second filling tube **130** is pivoted into the filling position, second casing brake unit **142** is shifted onto first end **132** of filling tube **130**, and the production of sausage-shaped products **S** may be continued. During the production process, a further tubular packaging casing supply **M** is transferred towards first filling tube **110** now in the refill position, and is stored thereon.

**[0238]** Refill of tubular packaging casing supply **M** onto second filling tube **130** arranged in the refill position, is executed by robotic device **300**, which takes up a tubular packaging casing supply **M** from first storage assembly **210** of storage device **200**, being in the unloading position, and transfers this tubular packaging casing supply **M** onto second filling tube **130** being arranged in the refill position.

**[0239]** Any position of robotic device **300**, like the take-up position, the allocation position and the release position as well as the moving paths of robotic device **300** between these positions are stored in control unit **RCU** of robotic device **300**, e.g. in the form of coordinates referring to a respective coordinate system. The moving path of robotic device **300** may be programmed by an operator and transferred to control unit **RCU** of robotic device **300**, or may directly be programmed into control unit **RCU** via a respective interface. Alternatively, robotic device **300** may manually be guided by an operator along a desired moving path that in the meantime is stored in control unit **RCU**. In all these cases, the movement of robotic device **300** is controlled by control unit **RCU**. Alternatively, the positions and the moving path of robotic device **300** may be stored in control unit **CU** of clipping machine **CM**, such that control unit **CU** of clipping machine **CM** may also control the movement of robotic device **300**.

**[0240]** For refilling tubular packaging casing supply **M** onto second filling tube **130** arranged in the refill position, robotic device **300** is moved into the take-up position (cf. FIG. 3). In the take-up position, first and second gripper elements **346**, **348** of gripper unit **340**, which are in their opened position, are aligned parallel and laterally to tubular packaging casing supply **M** stored on receiving pin **212** of first storage assembly **210** in the provision position. Gripper unit **340** is aligned to receiving pin **212** such that the first end **340a** of gripper unit **340** and the upper end of receiving pin **212** are at least approximately arranged at the same height. In this position, gripping axis **350**, at least approximately, coincides with the central axis of receiving pin **212**. Then, first and second drive assemblies **343**, **344** are activated such

that first and second gripper elements **346**, **348** are moved into their closed position for grasping tubular packaging casing supply M.

[0241] Each of tubular packaging casing supplies M arranged on receiving pins **212** of first storage assembly **210**, or on receiving pins **232** of second storage assembly **230**, respectively, is aligned with its upper end, which may already be closed by a closure clip C, to the upper end of respective receiving pins **212**, **232**. In this position, retaining elements **218** of first storage assembly **210**, and retaining elements **238** of second storage assembly **230**, respectively, are laid-up with pretension on the upper ends of receiving pins **212**, **232**, and thereby, abut on the upper ends of tubular packaging casing supplies M stored thereon. Retaining elements **218**, **238** thereby prevent tubular packaging casing supply M stored on receiving pins **212**, **232** from unintentional unfolding and exceeding upwardly from receiving pins **212**, **232** of first and second storage assemblies **210**, **230**.

[0242] For removing tubular packaging casing supply M from receiving pin **212**, robotic device **300** moves gripper unit **340** vertically upwards along receiving pin **212**, and about a length that corresponds to the length of base plate **342** of gripper unit **340**. Thereby, it is ensured that tubular packaging casing M is completely removed from receiving pin **212**, and collisions between gripper unit **340** and first storage assembly **210** are securely prevented. The tension applied to the upper end of tubular packaging casing supply M by retaining elements **218**, **238** is selected such that tubular packaging casing supply M is securely held on retaining pins **212**, **232**, but may be removed from receiving pins **212**, **232** without being damaged by retaining elements **218**, **238**, which are pushed or pivoted aside or upwards, preferably by gripper unit **340**.

[0243] Robotic device **300** is then moved into the allocation position as shown in FIG. 4, in which gripper unit **340** is arranged at least approximately horizontally on the rear side of clipping machine CM. In this position, tubular packaging casing supply M is held ready close to the refill position of filling tube **130**, whereby the time between the arrangement of second filling tube **130** in the refill position and the placement of tubular packaging casing supply M on second filling tube **130** is shortened, and possible downtimes of the production process are also shortened or prevented.

[0244] Second filling tube **130** is positioned in the refill position by rotating filling tube assembly **100** about rotation axis **154** of revolver plate **150**. Second casing brake assembly **140**, which is associated with second filling tube **130** such that second filling tube **130** and second casing brake assembly **140** are moved together towards the refill position, is in a position, in which second casing brake unit **142** is removed from first end **132** of filling tube **130**, by elongating telescopic casing brake holder unit **144** and pivoting second casing brake unit **142** together with casing brake holder unit **144** towards rotation axis **154** of revolver plate **150**.

[0245] When second filling tube **130** is arranged in the refill position, or a certain time period before, robotic device **300** is moved towards the refill position of second filling tube **130**. Gripper unit **340** is positioned with its second end **340b** immediately in front of first end **132** of second filling tube **130**, with gripping axis **350** being coaxially aligned with central axis A of second filling tube **130**.

[0246] Robotic device **300** is then moved into the release position by shifting gripper unit **340** along central axis A of

filling tube **130**, whereby tubular packaging casing supply M is transferred onto second filling tube **130**. Tubular packaging casing supply M is shifted onto second filling tube **130** such that its end that is closed by a closure clip C, which is the upper end when arranged on receiving pins **212**, **232** of first and second storage assemblies **210**, **230**, abuts first end **132** of second filling tube **130**.

[0247] Then, second casing brake assembly **140** is activated to pivot back towards second filling tube **130**, and to position casing brake unit **142** in front of first end **132** of second filling tube **130**.

[0248] Gripper unit **340** is moved into its opened position for releasing tubular packaging casing supply M onto second filling tube **130**. Second casing brake unit **142** positioned in front of first end **132** of second filling tube **130** prevents tubular packaging casing supply M from unintentional unfolding and falling off from second filling tube **130**.

[0249] Subsequently, robotic device **300** is moved away from second filling tube **130** and back into the take-up position for grasping a further tubular packaging casing supply M provided on a further receiving pin **212** arranged in the provision position. Second casing brake unit **142** is moved onto first end **132** of second filling tube **130** by retracting telescopic casing brake holder unit **144**, whereby tubular packaging casing supply M is finally secured on second filling tube **130**.

[0250] In order to continue the production of sausage-shaped products S, filling tube assembly **100** is rotated about rotation axis **154** of revolver plate **150**, such that first filling tube **110** is transferred into the refill position, and second filling tube **130** is moved into the lateral position in front of clipping machine CM. Second filling tube **130** with tubular packaging casing supply M thereon is pivoted into the filling position, and the production of sausage-shaped products S may be continued.

[0251] Refill of tubular packaging casing supply M onto first filling tube **110** is executed in the same manner as described in conjunction with second filling tube **130**.

[0252] It has to be noted that filling tube assembly **100**, for moving first and second filling tubes **110**, **130** from the lateral position into the refill position and vice versa, may be rotated stepwise about an angle of approximately 180° in the same direction, i.e. clockwise or counter clockwise, such that each filling tube **110**, **130** rotates about an angle of 360°, or reversibly about an angle of at least approximately 180°.

[0253] For enabling a correct grasping of tubular packaging casing supply M by gripper unit **340**, and for exactly position tubular packaging casing supply M on first and second filling tubes **110**, **130**, the respective positions and movement paths of robotic device **300** and gripper unit **340** may be programmed and implemented into a respective control logic of robotic device **300** in control unit RCU of robotic device **300** or in the control unit CU of clipping machine CM.

[0254] In order to ensure a safe and reliable production process on system **1000**, and to further ensure a safe refill operation, a first sensor unit SU1 is provided to gripper unit **340** of robotic device **300** (cf. FIG. 9). First sensor unit SU1 is arranged in the region first end **340a** of gripper unit **340** such that it faces towards gripper elements **346**, **348**.

[0255] First sensor unit SU1 is adapted to detect the position of gripper unit **340** relative to tubular packaging casing supply M provided on pin **212** of first storage assembly **210** in the provision position, during moving

robotic device **300** into the take-up position. First sensor unit **SU1** may include a suitable sensor element, like an optical sensor or a proximity sensor. Furthermore, for detecting or verifying the orientation of gripper unit **340**, first sensor unit **SU1** may also include a position sensor.

**[0256]** In the same way, first sensor unit **SU1** may also be used for detecting the position of gripper unit **340** relative to first or second filling tube **110**, **130** in the refill position, for securing a correct transfer of tubular packaging casing supply **M** onto first or second filling tube **110**; **130**.

**[0257]** A second sensor unit **SU2** may be provided, which includes a sensor element for detecting the gripping force exerted to tubular packaging casing supply **M** by first and second gripper elements **346**, **348**. Said sensor element may be a force sensor coupled to first and/or second drive assemblies **343**, **344** of gripper unit **340**. Second sensor unit **SU2**, particularly, its force sensor, may be integrated in first and/or second drive assemblies **343**, **344**, or may be attached to gripper elements **346**, **348**.

**[0258]** Furthermore, at least first and second storage assemblies **210**, **230** may be provided with a third sensor unit **SU3** in the region of their rotation plates **214**, **234**. Said sensor units **SU3** may detect the presence or absence of tubular packaging casing supply **M** on receiving pins **212**, **232**, and particularly on receiving pin **212**, **232** that is positioned in the provision position, as well as the rotation position of first and second storage assemblies **210**, **230**. Thereby, it may be ensured that tubular packaging casing supply **M** is hold ready in the provision position, for being grasped by gripper unit **340** of robotic device **300**, when in the take-up position. Third sensor unit **SU3** may integrally be formed with central posts **216**, **236**, or receiving pins **212**, **232**, and may include a respective sensor element, like a weight sensor or an optical sensor. Alternatively, third sensor unit **SU3** may also be provided on clipping machine **CM** in close vicinity to the provision position of first or second storage assembly **210**, **230** when in the unloading position, whereby only one third sensor unit **SU3** is necessary, and whereby the control amount is reduced and the design of the system is less complex.

**[0259]** It has to be noted, that in the FIGS. **2** to **11**, only possible positions of sensor units **SU1**, **SU2**, **SU3**, **SU4** are shown. A specific sensor element or a detailed embodiment of said sensor units **SU1**, **SU2**, **SU3**, **SU4** is not shown, since the respective sensors may integrally be arranged in the respective units or assemblies, like in first and second drive assemblies **343**, **344**, in central posts **216**, **236**, receiving pins **212**, **232** or support element **434**, and thus, sensor units **SU1**, **SU2**, **SU3**, **SU4** are not visible.

**[0260]** The control operation during the production of sausage-shaped products **S** in system **1000** may be executed as follows:

**[0261]** The production of sausage-shaped products may be started by selecting a specific sausage-shaped product **S** to be produced, e.g. by an operator, which selects said sausage-shaped product **S**, the data of which together with respective control and operation parameters may be stored in control unit **CM** of clipping machine **CM**. Alternatively, these information may separately be input by the operator into control unit **CU**.

**[0262]** On first filling tube **110** being in the filling position, a tubular packaging casing supply **M** is stored, and which is closed at its front end by a closure clip **C**. Sausage-shaped products **S** may now be produced, as explained in conjunc-

tion with FIG. **1**, by feeding filling material into tubular packaging casing **M** which is thereby pulled off from first filling tube **130**, gathering the filled tubular packaging casing **M**, forming a plait-like portion thereto and placing and closing at least one closure clip **C** thereto, for closing the filled portion of tubular packaging casing **M**. At the same time, a second closure clip **C** may be placed and closed on the plait-like portion, for closing the front end of the sausage-shaped product **S** subsequently to be produced. Additionally, a suspension element **L** is attached to sausage-shaped product **S** and fixed thereto by said at least one closure clips **C**. Sausage-shaped product **S** is severed from the remaining tubular packaging casing supply **M** by cutting the plait-like portion between the two closure clips **C**.

**[0263]** Suspension element **L** is caught by catching device **540** and guided towards spindle **410** of infed device **400**, while sausage-shaped product **S** is discharged from clipping machine **CM** by discharge device **DC**.

**[0264]** Sausage-shaped product **S**, hanging on spindle **410** by suspension element **L**, is transferred along spindle **410** onto guide bar **430**. A transportation element **540** of transport device **500** grasps suspension element **L** by its hooks **542** and guides sausage-shaped product **S** along guide bar **430** and rod-like element **R**. At a predefined position on rod-like element **R**, suspension element **L** is released from hooks **542**, and sausage-shaped product **S** is stored on rod-like element **R**.

**[0265]** While suspension element **L** of sausage-shaped product **S**, transported by spindle **410** or transportation element **540**, is passing fourth sensor unit **SU4** arranged either in the region of downstream end of spindle **410** or in the region of support element **434**, sensor unit **SU4** detects the presence of sausage-shaped product **S**, e.g. in conjunction with the passage of a transportation element **540**, and outputs a respective signal to control unit **HCU** of hanging line **HL**. This signal of fourth sensor unit **SU4** may be used for continuing the production process. In case that no sausage-shaped product **S** is detected by fourth sensor unit **SU4**, e.g. while an empty transportation element is passing fourth sensor unit **SU4**, a respective signal is sent to control unit **HCU** of hanging line **HL**, which in turn may act on transport device **500**. Alternatively or additionally, said sensor signal may be sent to control unit **CU** of clipping machine **CM**, which e.g. may stop hanging line **HL**.

**[0266]** The control operation during refilling tubular packaging casing supply **M** onto filling tube **130** may be executed as follows:

**[0267]** When starting the production of sausage-shaped products **S**, control unit **CU** of clipping machine **CM** initially activates robotic device **300**, e.g. for starting refill of tubular packaging casing supply **M** onto second filling tube **130** in the refill position, in the case that second filling tube **130** is not provided with a tubular packaging casing supply **M**. Said activation signal may be output to robotic device **300** on the basis of a signal received from a sensor unit (not shown) that is coupled to filling tube assembly **100**. Said sensor unit may output a signal corresponding to a specific position of filling tube assembly **100**, like a position in which second filling tube **130** while moving towards the refill position, is in a predefined angular position relative to the refill position.

**[0268]** Upon receipt of said signal of the sensor unit coupled to filling tube assembly **100**, robotic device **300** is activated. Robotic device **300**, being in the allocation position, is moved into a position as shown in FIG. **5**, in which

gripper unit **340** is positioned in front of first end **132** of second filling tube **130**, with gripping axis **350** of gripper unit **340** being coaxially aligned with central axis **A** of second filling tube **130** meanwhile being in the refill position. Sensor unit **SU1** detects the position of gripper unit **340** relative to second filling tube **130**. In the case that gripper unit **340** is correctly positioned in front of second filling tube **130**, a signal is provided by sensor unit **SU1** to control unit **RCU** of robotic device **300**, causing robotic device **300** to shift gripper unit **340** along the central axis of second filling tube **130**, such that tubular packaging casing supply **M** is transferred onto second filling tube **130**. While shifting tubular packaging casing supply **M** onto second filling tube **130**, first sensor unit **SU1** may further detect the position of first end **340a** of gripper unit **340** relative to second filling tube **130**, and may stop robotic device **300** when first end **340a**, and thus, the end of tubular packaging casing supply **M**, respectively, has reached first end **132** of second filling tube **130**. Alternatively or additionally, this position may be defined in the programming of robotic device **300**, which automatically stops in this position.

[0269] The signal of sensor unit **SU1**, indicating gripper unit **340** in the correct position on filling tube **130**, may also be used to activate second casing brake assembly **140** being in the position shown in FIG. 6, to pivot towards filling tube **130**, and to move second casing brake unit **142** towards first end **132** of second filling tube **130** into a position shown in FIG. 8. In this position, second casing brake unit **142** is arranged in front of first end **132** of second filling tube **130**.

[0270] Subsequently, first and second drive assemblies **343**, **344** of gripper unit **340** are activated to move gripper elements **346**, **348** into their opened position, to release tubular packaging casing supply **M** onto second filling tube **130**. Tubular packaging casing supply **M** is secured on filling tube **130** against unintentional unfolding by second casing brake unit **142**. Robotic device **300** may then be moved towards the take-up position or any intermediate position between the release position and the take-up position.

[0271] Second sensor unit **SU2** may confirm that tubular packaging casing supply **M** has been released onto second filling tube **130**, e.g. since no further force is applied to tubular packaging casing supply **M** by gripper elements **346**, **348** and/or gripper elements **346**, **348** are in a predefined position, and gripper unit **340** may be removed from filling tube **130**.

[0272] Thereafter, second casing brake unit **142** is shifted onto first end **132** of second filling tube **130** by retracting second telescopic casing brake holder unit **144** about a certain length, whereby the refill process is completed. A signal for activating second casing brake assembly **140** may be received from first sensor unit **SU1** that detects a respective position of gripper unit **340** relative to second filling tube **130** after being removed from second filling tube **130**.

[0273] Upon a respective signal indicating that tubular packaging casing supply **M** on first filling tube **110** arranged in the filling position is used up, filling tube assembly **100** is rotated about rotation axis **154** controlled by control unit **CU** of clipping machine **CM**, for moving second filling tube **130** with tubular packaging casing supply **M** stored thereon, toward the lateral position, and the filling position, respectively.

[0274] In the meantime, a receiving pin **212** of first storage assembly **210** with a tubular packaging casing supply **M** stored thereon, is positioned in the provision position. Third

sensor unit **SU3** detects the presence of tubular packaging casing supply **M** on said receiving pin **212**, and outputs a respective signal to control unit **CU** of clipping machine **CM**, or alternatively to control unit **RCU** of robotic device **300**. In the case that no tubular packaging casing supply **M** is stored on said receiving pin **212**, sensor unit **SU3** may cause first storage assembly **210** to rotate such that the next receiving pin **212** is moved into the provision position. These steps may be executed until a receiving pin **212** with a tubular packaging casing supply **M** stored thereon, is in the provision position. Alternatively, robotic device **300** may be stopped, and a respective signal may be sent to an operator. Furthermore, in the case that no receiving pin **212** of first storage assembly is provided with a tubular packaging casing supply **M**, rotation assembly **250** may be activated to rotate storage device **200** about rotation axis **252** and about an angle of approximately  $180^\circ$ , to move first storage assembly **210** from the unloading position into the loading position, in which tubular packaging casing supply **M** is loaded onto receiving pins **212**. Thereby, second storage assembly **230** is moved from the loading position into the unloading position, in which one of its receiving pins **232** may be placed in the provision position.

[0275] When arriving at the take-up position, first sensor unit **SU1** of gripper unit **340** detects receiving pin **212** in the provision position and the position of gripper unit **340** relative to said receiving pin **212**. Robotic device **300** positions gripper unit **340** such that gripper elements **346**, **348** are positioned laterally and parallel to receiving pin **212** with tubular packaging casing supply **M** thereon in the provision position. Gripping axis **350** is aligned coaxially with the central axis of receiving pin **212**. Gripper elements **346**, **348** are then moved towards each other into their closed position by first and second drive assemblies **343**, **344**, for grasping tubular packaging casing supply **M** stored on receiving pin **212**. Second sensor unit **SU2** detects the closing force exerted by gripper elements **346**, **348** onto tubular packaging casing supply **M**. After a predetermined force value is reached, first and second drive assemblies **343**, **344** are stopped. Gripper unit **340** is then moved upwardly along receiving pin **212** until gripper unit **340** is positioned with its second end **340b** above the upper end of receiving pin **212**. Thereafter, robotic device **300** is moved into the allocation position, from which the refill operation for refilling tubular packaging casing supply **M** onto first filling tube **110** now in the refill position, is executed in the same manner as described above in conjunction with second filling tube **130**.

[0276] As disclosed above, the moving paths between the positions of robotic device **300** may fixedly be programmed according to selected conditions, e.g. such that robotic device **300** is moved on the shortest path and/or in the shortest time. Alternatively, or additionally, robotic device **300** may be provided with additional sensors, like proximity sensors, that may sense the surrounding area of robotic device **300**, for detecting possible barriers on the moving path of robotic device **300**, like parts of clipping machine **CM** or other peripheral machines, and for preventing collisions therewith. Further additionally, the information provided by these additional sensors may be used to optimize the moving path of robotic device **300**, by an operator that varies the moving path or by self-learning of robotic device **300**.

[0277] Furthermore, alternative moving paths and/or positions of robotic device **300** may be stored in control unit RCU of robotic device **300** or in control unit CU of clipping machine CM. Said alternative moving paths and/or positions of robotic device **300** may be used for controlling clipping machine CM and robotic device **300** dependent on the kind of sausage-shaped products S to be produced, e.g. with regard to a varied casing material consumption, or dependent on the kind of peripheral machines used, like alternative filling tube assemblies or storage devices. Gripper unit **340** is replaceable mounted to second end **334** of second arm **330** of robotic device **300**. Thus, replacing gripper unit **340** by an alternative gripper unit, e.g. in adaption to the size of tubular packaging casing supply M, it may also be necessary to vary the positions and/or the moving path of robotic device **300**.

[0278] Moreover, sensor units SU1, SU2, SU3, SU4 and any additional sensor unit may be coupled to control unit CU of clipping machine CM, for activating clipping machine CM and/or robotic device **300** and/or hanging line HL, or for deactivating clipping machine CM and/or robotic device **600** and/or hanging line HL, e.g. in case of missing tubular packaging casing supply M or possible wrong or incorrect positioning of components of system **1000**. However, it is also possible that one or more of sensor units SU1, SU2, SU3, SU4 and any additional sensor unit are coupled to control unit RCU of robotic device **300** and or control unit HCU of hanging line HL, for activating or deactivating robotic device **300** and/or hanging line HL if necessary.

[0279] As explained above, robotic device **300** may be placed in the allocation position, to hold ready a tubular packaging casing supply M close to the refill position for shorten the refill time. However, dependent on the size of tubular packaging casing supply M and/or the time in which said tubular packaging casing supply M is used up, an allocation position may be omitted, and robotic device **300** may directly be moved from the take-up position into the release position.

[0280] Furthermore, in conjunction with FIGS. 2 to 10, filling tube assembly **100** has been described as including two filling tubes **110**, **130**. It has to be noted that the process of refilling tubular packaging casing supply onto a filling tube as disclosed above, may also be executed in a system with a filling tube assembly that includes only one filling tube, or that has more than two filling tubes, like three or four filling tubes. In such a case, only the angle of rotation for transferring a filling tube into the refill position varies, whereas the remaining steps and control operations may be executed as described above.

[0281] It is further possible that storage device **200**, even if not described above, includes a control unit for controlling the basic operations of storage device **200**, like the rotational movement of first and second storage assemblies **21**, **230** or rotation assembly **250**. However, also the control unit of storage device **200** may be coupled to control units CU, RCU of clipping machine CM or robotic device **300**, for at least partially be controlled by clipping machine CM or robotic device **300**, like starting or stopping storage device **200**.

[0282] Moreover, filling tube assembly **100** has been described as including a feeding socket **160** via which a feeding device, like a filler, for feeding filling material, like sausage meat, may be coupled to clipping machine CM. Said feeding device may also be coupled to system **1000**, particularly to control unit CU of clipping machine CM.

Accordingly, system **1000** may also control the operation of said feeding device. Additionally, a further sensor unit may be provided, for monitoring the feeding device, like the flow of filling material fed to filling tubes **110**, **130** of filling tube assembly **100**. Said sensor unit may be coupled to system **1000**, like to control unit CU of clipping machine CM, and may be arranged in the region of feeding socket **160** or at a suitable position at the feeding device. Naturally, also said feeding device may include a control unit that may be coupled to control unit CU of clipping machine CM. Thereby, also the feeding device may become part of system **1000**.

#### REFERENCE SIGNS

|        |                                                |
|--------|------------------------------------------------|
| [0283] | 10 filling tube                                |
| [0284] | 12 first end of the filling tube               |
| [0285] | 14 second end of the filling tube              |
| [0286] | 16 casing brake assembly                       |
| [0287] | 20 clipping device                             |
| [0288] | 22 first clipping tool                         |
| [0289] | 24 second clipping tool                        |
| [0290] | 30 gathering means                             |
| [0291] | 32 first displacer unit                        |
| [0292] | 34 second displacer unit                       |
| [0293] | 40 cutting device                              |
| [0294] | 50 discharge device                            |
| [0295] | 100 filling tube assembly                      |
| [0296] | 110 first filling tube                         |
| [0297] | 112 first end of first filling tube            |
| [0298] | 114 second end of first filling tube           |
| [0299] | 120 first casing brake assembly                |
| [0300] | 122 first casing brake unit                    |
| [0301] | 124 first telescopic casing brake holder unit  |
| [0302] | 130 second filling tube                        |
| [0303] | 132 first end of second filling tube           |
| [0304] | 134 second end of second filling tube          |
| [0305] | 140 second casing brake assembly               |
| [0306] | 142 second casing brake unit                   |
| [0307] | 144 second telescopic casing brake holder unit |
| [0308] | 150 revolver plate                             |
| [0309] | 152 drive device                               |
| [0310] | 154 rotation axis                              |
| [0311] | 200 storage device                             |
| [0312] | 210 first storage assembly                     |
| [0313] | 212 receiving pins                             |
| [0314] | 214 rotation plate                             |
| [0315] | 216 central post                               |
| [0316] | 218 retaining elements                         |
| [0317] | 230 second storage assembly                    |
| [0318] | 220 rotation axis                              |
| [0319] | 222 rotary drive                               |
| [0320] | 232 receiving pins                             |
| [0321] | 234 rotation plate                             |
| [0322] | 236 central post                               |
| [0323] | 238 retaining elements                         |
| [0324] | 240 rotation axis                              |
| [0325] | 242 rotary drive                               |
| [0326] | 250 rotation assembly                          |
| [0327] | 252 rotation axis                              |
| [0328] | 254 rotation table                             |
| [0329] | 300 robotic device                             |
| [0330] | 310 base                                       |
| [0331] | 320 first arm                                  |
| [0332] | 322 first end of first arm                     |



[0333] 324 second end of first arm  
 [0334] 330 second arm  
 [0335] 332 first end of second arm  
 [0336] 334 second end of second arm  
 [0337] 340 gripper unit  
 [0338] 340a first end of the gripper unit  
 [0339] 340b second end of the gripper unit  
 [0340] 342 base plate  
 [0341] 342a first end of the base plate  
 [0342] 342b second end of the base plate  
 [0343] 343 first drive assembly  
 [0344] 344 second drive assembly  
 [0345] 345 connecting unit  
 [0346] 346 first gripper element  
 [0347] 346a, 346b brackets  
 [0348] 348 second gripper element  
 [0349] 348a, 348b brackets  
 [0350] 350 gripping axis  
 [0351] 400 infeed device  
 [0352] 410 spindle  
 [0353] 412 toothed surface  
 [0354] 420 spindle bearing  
 [0355] 422 support roller  
 [0356] 424 toothed portion  
 [0357] 430 guide bar  
 [0358] 432 grooves  
 [0359] 434 support element  
 [0360] 436 engaging holes  
 [0361] 440 connecting element  
 [0362] 442 grooves  
 [0363] 450 catching needle  
 [0364] 452 first end  
 [0365] 454 second end  
 [0366] 456 central portion  
 [0367] 500 transport device  
 [0368] 510 conveyor chains  
 [0369] 520 deflection rollers  
 [0370] 530 common axle  
 [0371] 540 transportation element  
 [0372] 542 hooks  
 [0373] 1000 system for producing sausage-shaped products  
 [0374] A central axis of filling tubes  
 [0375] C closure clip  
 [0376] CD clipping device  
 [0377] CM clipping machine  
 [0378] CU control unit of the clipping machine  
 [0379] DC discharge device  
 [0380] F feeding direction  
 [0381] GM gathering means  
 [0382] HL hanging line  
 [0383] HCU control unit of the hanging line  
 [0384] ID infeed direction  
 [0385] L suspension element  
 [0386] M tubular packaging casing/tubular packaging casing supply  
 [0387] P plait-like portion  
 [0388] RCU control unit of the robotic device  
 [0389] S sausage-shaped product  
 [0390] SC supply of closure clips  
 [0391] SU1 first sensor unit

[0392] SU2 second sensor unit  
 [0393] SU3 third sensor unit  
 [0394] SU4 fourth sensor unit  
 [0395] T transportation direction

1. A system for producing sausage-shaped products, like sausages, by filling a filling material into a tubular packaging casing, the system includes a clipping machine for producing sausage-shaped products, a robotic device for transferring a tubular packaging casing supply to the clipping machine, and a storage device for storing a tubular packaging casing supply,

the clipping machine comprises:

a filling tube assembly having at least a first filling tube for feeding the filling material into the tubular packaging casing stored on the filling tube and closed at its first end,

at least a first casing brake assembly, associated with the first filling tube, for applying a braking force to the tubular packaging casing while being pulled-off from the first filling tube, and

a control unit for controlling at least the clipping machine,

the robotic device includes:

a gripper unit for gripping a tubular packaging casing supply provided by the storage device, and for transferring said tubular packaging casing supply onto the at least first filling tube of the clipping machine;

the storage device includes:

at least a first storage assembly in which at least one tubular packaging casing supply is provided for being transferred by the robotic device onto the at least first filling tube of the clipping machine.

2. The system according to claim 1, wherein the control unit of the clipping machine is adapted to at least activate the robotic device for starting the process of refilling tubular packaging casing supply onto the first filling tube.

3. The system according to claim 1, wherein the first filling tube is movable at least between a filling position, in which filling material can be fed through the first filling tube into the tubular packaging casing stored on the first filling tube, and a refill position, in which a tubular packaging casing supply can be transferred onto the first filling tube.

4. The system according to claim 1, wherein the filling tube assembly includes a second filling tube and a second casing brake assembly associated with the second filling tube, wherein, preferably, the first filling tube is arranged in the filling position while the second filling tube is arranged in the refill position.

5. The system according to claim 1, wherein the filling tube assembly further comprises a revolver plate, with at least the first filling tube attached to the revolver plate, the revolver plate being rotatable about a rotation axis for reversibly moving the at least first filling tube from the filling position into the refill position.

6. The system according to claim 1, wherein the robotic device includes a control unit, and wherein the control unit of the clipping machine and the control unit of the robotic device are interconnected such that the robotic device at least partially may be controlled by the control unit of the clipping machine and/or the clipping machine at least partially may be controlled by the robotic device.

7. The system according to claim 1, wherein the robotic device includes at least a first sensor unit at least for detecting the position of the gripper unit relative to the

storage device and/or the position of the gripper unit relative to the first and/or second filling tube arranged in the refill position.

8. The system according to claim 1, wherein the robotic device includes a second sensor unit for determining a gripping force applied to the tubular packaging casing supply by the gripper unit.

9. The system (1000) according to claim 1, wherein the storage device includes at least a third sensor unit for detecting the presence or absence of a tubular packaging casing supply in the storage device.

10. The system according to claim 1, further including a hanging line arranged downstream the discharge device, for suspending the sausage-shaped products discharged from the clipping machine by the discharge device onto a rod-like element, like a smoking rod, the hanging line includes a control unit for controlling at least the hanging line.

11. The system according to claim 1, wherein the control unit of the hanging line is interconnected with the control unit of the robotic device and/or the control unit of the clipping machine, such that the hanging line is at least partially controlled by the clipping machine and/or the robotic device.

12. The system according to claim 10, wherein the hanging line includes at least one fourth sensor unit for detecting the presence or absence of a sausage-shaped product in the hanging line.

13. A method for producing sausage-shaped products in a system according to claim 1, the method comprises the steps of:

feeding a filling material through the at least first filling tube of the clipping machine into a tubular packaging casing stored on the at least first filling tube, gathering the filled tubular packaging casing by the gathering means of the clipping machine and forming a plait-like portion thereto, applying at least one closure means to the plait-like portion and closing said closure means by the clipping device of the clipping machine, discharging the sausage-shaped product just produced out of the clipping machine by the discharge device of the clipping machine,

the method further comprises the steps of: moving the at least first filling tube from the filling position, in which filling material is fed into the tubular packaging casing, into the refill position, in which a tubular packaging casing supply is transferred onto the at least first filling tube, and transferring by the robotic device a tubular packaging casing supply provided by a storage device onto the at least first filling tube being arranged in the refill position.

14. The method according to claim 13, further comprising the step of: suspending the sausage-shaped product discharged by the discharge device on a rod-like element provided by the hanging line.

15. The method according to claim 13, further comprising the step of: least partially controlling by the clipping machine the robotic device and/or the hanging line.

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